

Mission Test Final Report

BETA CREW

PHYC41110 – Spacecraft Operations

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1. Introduction

This report presents an analysis of the BETA crew mission test and the spacecraft operations carried out throughout the exercise. The mission test consisted of five RF communication passes with the EIRSAT-1 FlatSat system, during which team members rotated between several operational roles including Test Coordinator, Test Support Operator, Daily Pass Planner and Spacecraft Operations roles. The overall aim of the mission test was to simulate realistic spacecraft flight operations using real operational procedures, limited communications windows and telemetry handling under realistic mission constraints.

Each pass involved pre-pass preparation, real-time spacecraft operations during the communications window and post-pass analysis of telemetry and operational events. Although the scheduled pass durations were approximately ten minutes, the actual usable communications time between AOS and LOS varied between passes and created genuine operational pressure throughout the mission test. Activities performed during the passes included telemetry downlink, payload operations, operational mode management, telemetry analysis using Grafana dashboards and spacecraft recovery procedures following unexpected anomalies.

A major focus of the mission test was the downlink and analysis of EVENT, HOUSEKEEPING and EMOD telemetry. Large Data Transfer (LDT) files were successfully recovered from onboard storage and used to assess spacecraft and payload health throughout operations. EMOD telemetry was also decoded and analysed to evaluate payload performance against mission elapsed time. During Shift 2, the spacecraft entered safe mode following a significant battery voltage drop, creating a realistic contingency scenario that required rapid operator coordination, telemetry monitoring and spacecraft recovery procedures.

This report mainly reflects on my own experience during the mission test and evaluates both the operational success of the mission and the challenges encountered throughout spacecraft operations. It discusses the effectiveness of the simulation in replicating real spacecraft operations, the impact of RF communications and limited pass durations on operational difficulty, and the importance of preparation, communication and teamwork during live spacecraft operations.

2. Part A: Mission Test Experience

Mission Test

In my experience, the spacecraft operational procedure mission test was overall a very positive experience and I can almost say a success, despite all of the trials and tribulations along the way. The roles were broken down into several different responsibility categories. Please see 'Table 1' below for role allocation schedule of the mission test passes.

Table 1. Pass Schedule role allocation of BETA Crew Mission Test Passes.

Shift No.	**Pass No.**	**Pass Time**	SpaCom	Spacecraft Operator	Test Support Operator	Test Coordinator	Daily Pass Planner
1	1	11:45	AM	JA	LM	OND	ED
	2	12:45	LM	ED	OND	JA	AM
2	3	13:30	OND	AM	JA	ED	LM
	4	14:30	JA	LM	ED	AM	OND
	5	15:30	ED	OND	AM	LM	JA

Objectives of the mission test

The overall objectives of the mission test were to:

- Simulate realistic nominal and contingency main in-flight situations and spacecraft operations using simulated EIRSAT-1 data and operations room.
- Professionally execute scheduled passes, including pre-pass preparation, real-time operations during each pass, and post-pass analysis and documentation of events that occurred during passes.
- Downlink Large Data Transfer (LDT) files containing logged EMOD data from the satellite.
- Develop and use a decoder to process LDT data and graph EMOD temperatures as a function of mission elapsed time using the developed decoder.
- Assess the operational status and health of the EMOD payload using the decoded telemetry data.
- Perform necessary uplink operations within defined communication windows if it is required.
- Evaluate the impact of RF communications and ground station constraints on spacecraft operations.
- Validate and refine operational procedures (TS and TP) during mission execution.
- Use the mission test as an opportunity to develop operator familiarity with mission control systems, tools, and workflows.

The success of the mission test

Overall, the mission was a success, I wouldn't say it was perfect, but we did achieve 'most' of the mission objectives mentioned above. We did run into a major issue or 'curse' bestowed upon us from

our lovely support crew, the Alpha crew, but despite this it was still an overall success as we did achieve most of the objectives mentioned above. We did use the opportunity to realistically simulate a nominal and contingency in-flight situation and perform spacecraft operations during this using EIRSAT-1 data and operations room. We also successfully and professionally executed scheduled passes with pass preparation before each pass, real-time operations during each pass, and post-pass analysis. We downlinked EVENTS and HOUSKEEPING telemetry data files for all passes, and TED and EMOD during some passes too. We also succeeded in performing necessary uplinking operations where required, validated and refined our 'Test Procedure' and 'Test Specifications' documents. More than all of the above objectives we most definitely used the mission test as an opportunity to develop our personal familiarity with mission control systems, tools and workflows.

Were all the scenarios completed as planned?
Hmm… that's a tough question to answer… Honestly I think there were many opportunities to do better and complete scenarios better, that said this was our very first opportunity to work on something like this and to also be tested simultaneously was an additional stress that added to the experience. So, 'Success' in the metaphorical sense; yes. 'Success' in the 'did we complete all mission objectives and secenarios in their entirety; not this time… unfortunately!

Shifts

This mission test was split into two separate shifts. The first shift (RED) included two passes with the PASS 1 starting at 11:45:00 UTC and then PASS 2 at 12:45:00 UTC. The second shift (BLUE) included three passes, with the PASS 3 beginning at 14:30:00 UTC, PASS 4 at 15:30:00 UTC and the final pass of the day, PASS 5 starting at 16:30:00 UTC. My roles during both shifts are outlined below;

- Shift 1. Test Coordinator, Test Support Officer, SpaCom.
- Shift 2. Daily Pass Planner, Spacecraft Operator.

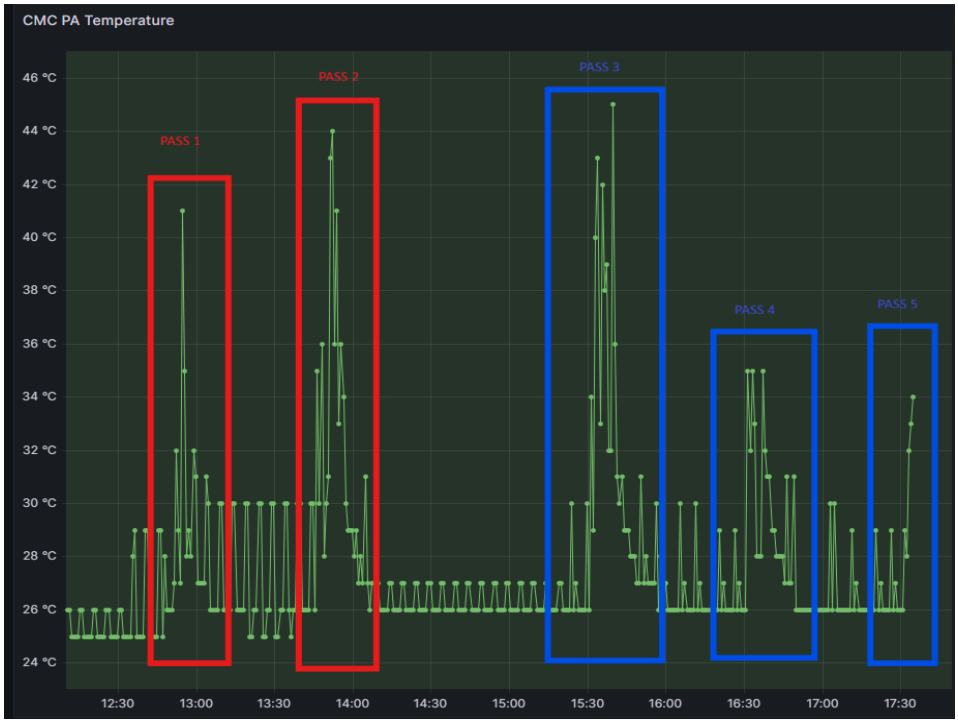


Figure 1. Shift and Passes as seen using Grafana Temperature Data

(Note Grafana set time is local to device which is currently UTC +2)

Planned Objectives of each shift (1) & (2);

1. The planned objectives for the first shift included priority of obtaining downlinks of EVENTS and HOUSEKEEPING telemetry to assess satellite health. We also had the secondary objective of obtaining TED and PASCAL telemetry if achievable. The reasoning behind this is because it was the first downlink in seven days and therefore, the data would be many rows and may take some time, so it was a primary goal of the first pass to ensure downlink of channel 8 and 15.
2. The planned objectives of the second shift was to programme and downlink data from the EMOD system onboard. With the final objective of ensuring EMOD is deactivated before the end of the final pass.

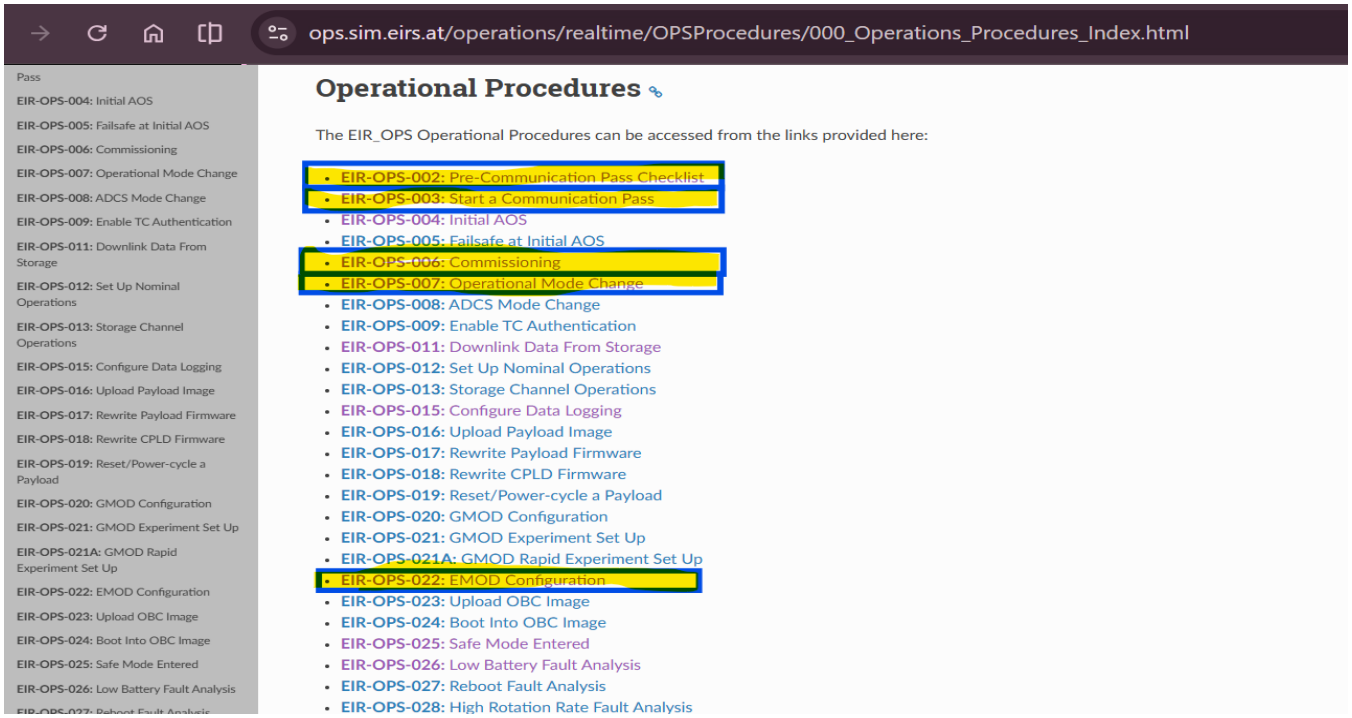


Figure 2. Operational Procedure Resources from https://ops.sim.eirs.at/operations/realtime/realtime.html#eir-ops

Shift assessment of the spacecraft and payload Health and Operational status.

SHIFT 1.

***Note: See all associated Grafana screenshots for this shift in section* 5. Appendix; SHIFT 1 (11:45:00 – 14:45:00 UTC+2) Grafana Screenshots**

The main aim of Shift 1 was to get communication with the spacecraft and downlink EVENT and HOUSEKEEPING telemetry in order to assess the overall spacecraft health and operational status. Since this was the first downlink in around a week, there was a large amount of stored telemetry onboard, meaning telemetry recovery had to be prioritised during the short communications windows.

Telemetry monitoring was mainly performed using the Grafana dashboards found in the appendix. The telemetry plots showed generally stable spacecraft behaviour throughout both passes, with EPS voltage, current and temperature telemetry remaining pretty stable overall. Some variation in current telemetry was visible during active downlink operations, which was expected during communications activity. Event telemetry also increased steadily throughout the passes, which means there was continuous spacecraft activity and successful telemetry downlinking.

During Pass 1, EVENT telemetry was successfully recovered while the HOUSEKEEPING downlink was only partially completed due to pass duration limitations, it went by much quicker than expected. Despite this, enough telemetry was received to confirm that the spacecraft was both operational and

responsive throughout the pass. During Pass 2, a way larger amount of housekeeping telemetry was successfully downlinked, providing a far more complete assessment of spacecraft subsystem behaviour. Overall, no major subsystem instability, critical anomalies or problems were observed during Shift 1 operations.



Figure 3. General health Grafana screenshots of Shift 1. (Including telemetry graphs of Mode Status, Event Counts, Uptime, OBC temperature, CMC 5V Voltage, CMC PA Temperatures).

SHIFT 2.

***Note: See all associated Grafana screenshots for this shift in section* 5. Appendix; SHIFT 2 (14:45:00 – 17:45:00 UTC+2) Grafana Screenshots**

Shift 2 focused mainly on EMOD payload operations, telemetry recovery and spacecraft configuration activities. Telemetry work was continued using the Grafana dashboards and downlinked telemetry files included in the Apendix. The third pass proved to be our most successful team pass as we achieved the mission objective of configuring EMOD, downliked both HOUSEKEEPING and EVENTS and even managed to obtain some TED (Telemetry-Enhanced Diagnostics) downlinks.

Overall, the spacecraft remained operational throughout the whole shift, although during the fourth pass with an AOS of 15:30 UTC the team ran into a big issue caused by an unexpected battery voltage drop. During this pass, telemetry showed the EPS battery voltage decreasing to approximately 7.41V, which resulted in the spacecraft entering safe mode. Despite this anomaly, communications with the spacecraft remained stable and the spacecraft continued responding to our input commands. The telemetry plots showed a very noticeable change in power and current telemetry during this period, which proved the spacecrafts transition into safe mode and the associated operational changes.

During the final pass, the spacecraft was successfully recovered from safe mode back into commissioning mode, which allowed us to obtain EVENT, HOUSEKEEPING and some EMOD telemetry

to be successfully downlinked. The successful recovery of EMOD telemetry confirmed that the payload remained operational during the final communications pass. Overall, Shift 2 successfully demonstrated spacecraft recovery procedures, continued telemetry acquisition and successful payload operations despite the temporary safe mode event. Unfortunately during this pass we were limited by the pass time constraints and unfortunately could not achieve the final shift objective of deactivating EMOD, though this may not be too impactful for future operations.

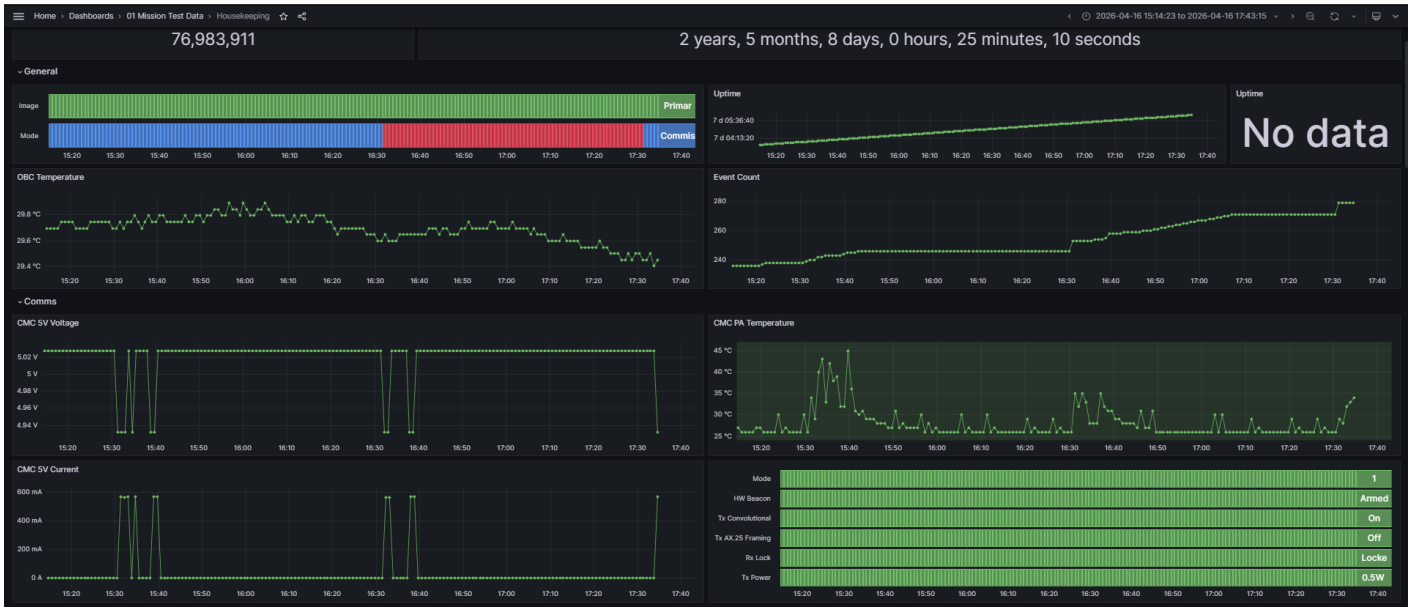


Figure 4. General health Grafana screenshots of Shift 1. (Including telemetry graphs of Mode Status, Event Counts, Uptime, OBC temperature, CMC 5V Voltage, CMC PA Temperatures).

Passes

This section details information on each pass including my role during the pass, any pre-pass preparations made before that pass, the pass length and expected objectives to be completed during that pass, all procedures and objectives successfully completed during that pass. It also includes details of difficulties encountered during each pass. Lastly, this section highlights any data that has been both downlinked and uplinked during each of the passes.

To avoid repetition, for all passes the following steps were taken during pre-pass preparation;

- Before each pass, the team reviewed the planned objectives, assigned operator roles and confirmed the expected spacecraft operational state prior to AOS.
- Telemetry priorities and planned downlink activities were discussed in advance based on shift objectives.
- The team also reviewed the relevant operational procedures from the EIR-OPS manual, prepared Grafana telemetry monitoring dashboards and ensured that communications systems, pass plans and mission control software were ready before the beginning of each communications window.

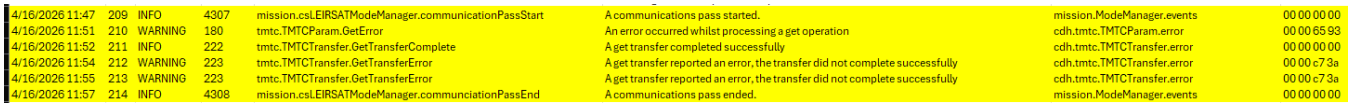
- In addition, the previous pass telemetry (if available) and spacecraft status were reviewed before proceeding with further operational activities.

Note: *All the associated data files listed in the 'Data Downlinked' rows within each pass table below can be found on this team Google Drive file storage link.*

Note: For full EIRSAT Manual procedures followed as referenced below please see the links provided in section **5. Appendix EIRSAT-1 Operations Manual Reference Procedures**

PASS 1; 11:45:00 - 11:55:00 UTC

My Role	Test Coordinator
Pre-Pass Preparation	Before the pass, the team reviewed the pass plan, confirmed operator roles and discussed telemetry downlink priorities. Since this was the first downlink in approximately one week, priority was given to downlinking EVENT and HOUSEKEEPING telemetry to assess the spacecraft health and EIRSAT's operational status. We also reviewed the expected spacecraft state, operational mode and communications procedures before AOS.
Specific Objectives of Pass	The main objective of Pass 1 was to start communications with and begin downlinking EVENT and HOUSEKEEPING telemetry from onboard storage. The secondary objectives of this pass was to begin assessment of spacecraft subsystem health using Grafana telemetry monitoring.
Pass Length	2026-04-16 11:47:07 - 2026-04-16 11:57:29 Duration; 10 minutes 22 seconds
Procedures Completed During Pass	The team completed the pre-comms pass initialisation procedure and successfully started the communications pass. EVENT telemetry was successfully downlinked and partial HOUSEKEEPING telemetry was recovered too. Telemetry monitoring was carried out throughout the pass using Grafana dashboards and the display of the waterfall plot. • **EIR-OPS-002: Pre-Communication Pass Checklist** • **EIR-OPS-003: Start a Communication Pass** • **EIR-OPS-011: Downlink Data From Storage**
Objectives Achieved	Communications with the spacecraft were successfully established and telemetry was successfully received. EVENT telemetry was fully recovered and partial HOUSEKEEPING telemetry was successfully downlinked, allowing an initial assessment of spacecraft health and operational status.
Difficulties Encountered	Due to the short communications window and the large amount of stored telemetry onboard, the housekeeping downlink could not be fully completed during the pass. Telemetry prioritisation was very important during operations of this first pass.
Data Downlinked	○ EVENT telemetry; `Event_20260416_1200.dat` ○ Partial HOUSEKEEPING telemetry; `HK_20260416_1200_Partial`
Data Uplinked	NA



4/16/2026 11:47	209	INFO	4307	mission.csl.EIRSATModeManager.communicationPassStart	A communications pass started.	mission.ModeManager.events	00 00 00 00
4/16/2026 11:51	210	WARNING	180	tmtc.TMTCParam.GetError	An error occurred whilst processing a get operation	cdh.tmtc.TMTCParam.error	00 00 65 93
4/16/2026 11:52	211	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 11:54	212	WARNING	223	tmtc.TMTCTransfer.GetTransferError	A get transfer reported an error, the transfer did not complete successfully	cdh.tmtc.TMTCTransfer.error	00 00 c7 3a
4/16/2026 11:55	213	WARNING	223	tmtc.TMTCTransfer.GetTransferError	A get transfer reported an error, the transfer did not complete successfully	cdh.tmtc.TMTCTransfer.error	00 00 c7 3a
4/16/2026 11:57	214	INFO	4308	mission.csl.EIRSATModeManager.communciationPassEnd	A communications pass ended.	mission.ModeManager.events	00 00 00 00

Figure 5. Event log extract for Pass 1 taken from full .csv-Excel file of 'EVENTS' data from full Mission Test Day.

PASS 2; 12:45:00 - 12:55:00 UTC

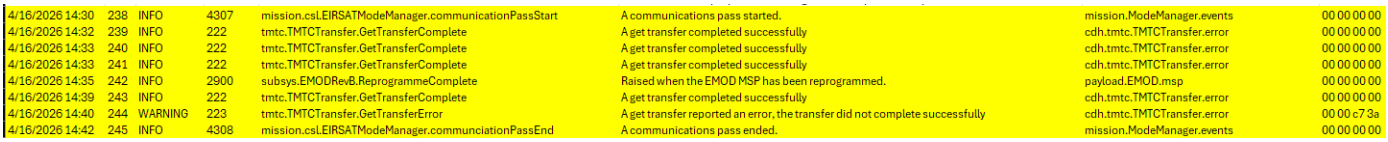
My Role	Test Support Officer
Pre-Pass Preparation	Before the pass, the team reviewed the results of Pass 1 and confirmed the remaining telemetry downlink priorities. Since the housekeeping downlink from Pass 1 was incomplete, recovering additional HOUSEKEEPING telemetry remained a main priority for Pass 2. Telemetry monitoring tools, Grafana dashboards and pass procedures were also prepared before AOS.
Specific Objectives of Pass	The main objective of Pass 2 was to continue downlinking EVENT and HOUSEKEEPING telemetry from onboard storage in order to further assess spacecraft health and check for any unexpected data.
Pass Length	2026-04-16 12:45:53 - 2026-04-16 12:57:19 Duration: 11 minutes 26 seconds
Procedures Completed During Pass	The communications pass initialisation procedure was completed successfully and telemetry downlink operations were carried out throughout the pass. EVENT and HOUSEKEEPING telemetry were downlinked as expected • **EIR-OPS-002: Pre-Communication Pass Checklist** • **EIR-OPS-003: Start a Communication Pass** • **EIR-OPS-011: Downlink Data From Storage**
Objectives Achieved	A significantly larger quantity of housekeeping telemetry was successfully downlinked compared to Pass 1, which allowed us to improve our assessment of the spacecraft subsystem behaviour and operational status. Telemetry monitoring confirmed continued spacecraft stability during operations.
Difficulties Encountered	The large amount of stored telemetry continued to create time pressure during the pass, requiring telemetry priorities to be done and we could not complete any further downlinks
Data Downlinked	○ EVENT telemetry; `Event_20260416_1300.dat` ○ HOUSEKEEPING telemetry; `HK_20260416_1300.dat`
Data Uplinked	NA

4/16/2026 12:45	215	INFO	4307	mission.csLEIRSATModeManager.communicationPassStart	A communications pass started.	mission.ModeManager.events	00 00 00 00
4/16/2026 12:47	216	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 02 30 12
4/16/2026 12:47	217	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:49	218	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 02 30 12
4/16/2026 12:49	219	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:50	220	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:51	221	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:52	222	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:52	223	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 12:53	224	WARNING	180	tmtc.TMTCParam.GetError	An error occurred whilst processing a get operation	cdh.tmtc.TMTCParam.error	00 03 57 7a
4/16/2026 12:54	225	WARNING	190	tmtc.TMTCAction.InvokeError	An error occurred whilst processing an invoke operation	cdh.tmtc.TMTCAction.error	00 00 00 02
4/16/2026 12:54	226	WARNING	190	tmtc.TMTCAction.InvokeError	An error occurred whilst processing an invoke operation	cdh.tmtc.TMTCAction.error	00 00 65 96
4/16/2026 12:55	227	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 02 30 12
4/16/2026 12:56	228	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 02 30 12
4/16/2026 12:57	229	INFO	4308	mission.csLEIRSATModeManager.communciationPassEnd	A communications pass ended.	mission.ModeManager.events	00 00 00 00

Figure 6. Event log extract for Pass 2 taken from full .csv-Excel file of 'EVENTS' data from full Mission Test Day.

PASS 3; 13:30:00 – 13:40:00 UTC

My Role	Space Commander
Pre-Pass Preparation	Before the pass, the team reviewed the telemetry recovered during Shift 1 and prepared for EMOD payload operations planned for Shift 2. The operational procedures for EMOD reprogramming, health checks and data handling were reviewed before AOS. The team also completed a test run of the operations to best prepare for this pass.
Specific Objectives of Pass	The main objective of Pass 3 was to begin EMOD payload operations, including EMOD configuration and preparation for payload telemetry collection. Secondary objectives included continued spacecraft health monitoring and potential TED downlinking.
Pass Length	2026-04-16 14:30:47 - 2026-04-16 14:42:29 11 minutes 42 seconds
Procedures Completed During Pass	The communications pass was successfully initialised and EMOD operational procedures were carried out during the pass. Telemetry monitoring continued throughout operations using Grafana and the waterfall plots and spacecraft health checks were performed before payload activity proceeded. • **EIR-OPS-002: Pre-Communication Pass Checklist** • **EIR-OPS-003: Start a Communication Pass** • **EIR-OPS-011: Downlink Data From Storage** • **EIR-OPS-022: EMOD Configuration**
Objectives Achieved	EMOD operational activities were successfully initiated and the spacecraft remained stable throughout the pass. Telemetry monitoring confirmed continued nominal spacecraft behaviour during payload operations.
Difficulties Encountered	This pass was overall very successful but time constraints meant that only partial TED downlinks were achieved.
Data Downlinked	○ EVENT telemetry; `EVENT_20260416_1442.dat` ○ HOUSEKEEPING telemetry; `HK_20260416_1442.dat` ○ Partial TED telemetry (Two Separate file chunks); `TED_20260416_1442_1.dat` , `TED_20260416_1442_2.dat`
Data Uplinked	○ EMOD configuration ○ Operational telecommands associated with payload preparation procedures.



4/16/2026 14:30	238	INFO	4307	mission.csl.EIRSATModeManager.communicationPassStart	A communications pass started.	mission.ModeManager.events	00 00 00 00
4/16/2026 14:32	239	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 14:33	240	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 14:33	241	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 14:35	242	INFO	2900	subsys.EMODRevB.ReprogrammeComplete	Raised when the EMOD MSP has been reprogrammed.	payload.EMOD.msp	00 00 00 00
4/16/2026 14:39	243	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 14:40	244	WARNING	223	tmtc.TMTCTransfer.GetTransferError	A get transfer reported an error, the transfer did not complete successfully	cdh.tmtc.TMTCTransfer.error	00 00 c7 3a
4/16/2026 14:42	245	INFO	4308	mission.csl.EIRSATModeManager.communciationPassEnd	A communications pass ended.	mission.ModeManager.events	00 00 00 00

Figure 7. Event log extract for Pass 3 taken from full .csv-Excel file of 'EVENTS' data from full Mission Test Day.

PASS 4; 14:30:00 – 14:40:00 UTC

My Role	Daily Pass Planner
Pre-Pass Preparation	Before the pass, the planned operational activities and telemetry priorities were reviewed with the team. Particular attention was given to EMOD payload operations, telemetry recovery and spacecraft power status due to the increasing operational workload during Shift 2.
Specific Objectives of Pass	The main objective of Pass 4 was to continue EMOD payload operations and telemetry downlink activities while monitoring spacecraft subsystem health throughout the pass.
Pass Length	2026-04-16 15:30:31 - 2026-04-16 15:42:19 Duration: 11 minutes 48 seconds
Procedures Completed During Pass	The communications pass was successfully initialised and telemetry monitoring was carried out. EMOD operational activities and telemetry downlink procedures continued during the pass until we encountered our voltage drop issue. • **EIR-OPS-002: Pre-Communication Pass Checklist** • **EIR-OPS-003: Start a Communication Pass** • **EIR-OPS-011: Downlink Data From Storage**
Objectives Achieved	Telemetry and payload operations continued successfully during the early part of the pass and spacecraft telemetry was successfully monitored throughout operations.
Difficulties Encountered	During the pass, the EPS battery voltage dropped significantly to approximately 7.41V, which automatically caused the spacecraft to enter into safe mode. This interrupted planned operations and required the team to focus on spacecraft recovery and operational monitoring for the remainder of the pass.
Data Downlinked	○ EVENT telemetry; `Event_20260416_1630.dat` ○ HOUSEKEEPING telemetry; `HK_20260416_1630.dat`
Data Uplinked	○ Operational telecommands associated with payload activities and spacecraft monitoring procedures.

4/16/2026 15:30	246	INFO	4307	mission.cslEIRSATModeManager.communicationPassStart	A communications pass started.	mission.ModeManager.events	00 00 00 00
4/16/2026 15:30	247	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 03 30 12
4/16/2026 15:30	248	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage threshold, where the default threshold is 7.5 V.	cdh.monitoring.CriticalMonitor.error	00 00 30 12
4/16/2026 15:30	249	WARNING	4306	mission.cslEIRSATModeManager.SafeModeTrigger	This event will be recorded when EIRSAT-1 is triggered to enter safe mode.	mission.ModeManager.events	00 00 00 07
4/16/2026 15:30	250	INFO	4301	mission.cslEIRSATModeManager.ExitModeSuccess	Record the successful exit of a specified mode.	mission.ModeManager.events	00 00 00 01
4/16/2026 15:30	251	INFO	319	mission.cslADCSManager.ModeStateSuccess	Raised when the ADCS mode and state are successfully set	mission.ADCSManager.adcs	00 00 00 00
4/16/2026 15:30	252	INFO	4302	mission.cslEIRSATModeManager.EnterModeSuccess	Record the successful entry to a specified mode.	mission.ModeManager.events	00 00 00 04
4/16/2026 15:35	253	WARNING	181	tmtc.TMTCParam.SetError	An error occurred whilst processing a set operation	cdh.tmtc.TMTCParam.error	00 00 00 02
4/16/2026 15:38	254	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 15:38	255	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes of setting TMBeacon.periodMultiplier.	cdh.monitoring.BeaconRateMonitor.error	00 03 30 12
4/16/2026 15:38	256	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage threshold, where the default threshold is 7.5 V.	cdh.monitoring.CriticalMonitor.error	00 00 30 12
4/16/2026 15:39	257	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 15:42	258	INFO	4308	mission.cslEIRSATModeManager.communciationPassEnd	A communications pass ended.	mission.ModeManager.events	00 00 00 00

Figure 8. Event log extract for Pass 4 taken from full .csv-Excel file of 'EVENTS' data from full Mission Test Day. ***Note; RED shows critical warning messages.***

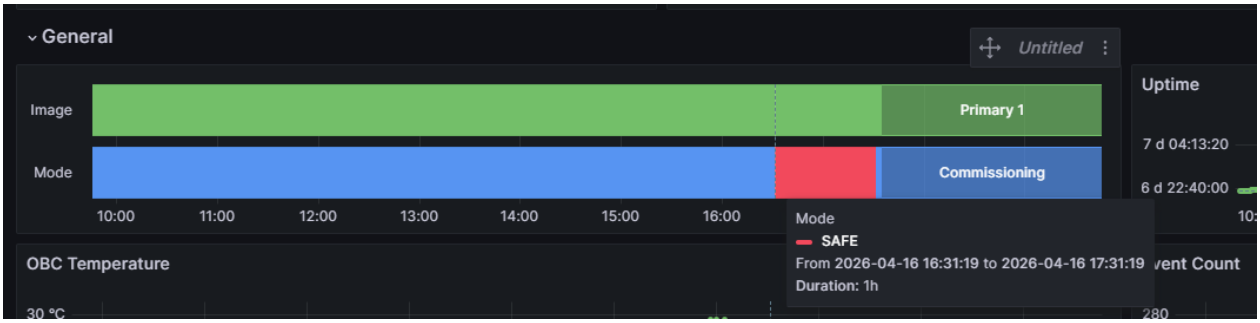


Figure 9. Screenshot from Grafana showing mode switch from 'Commissioning' to 'safe mode'.

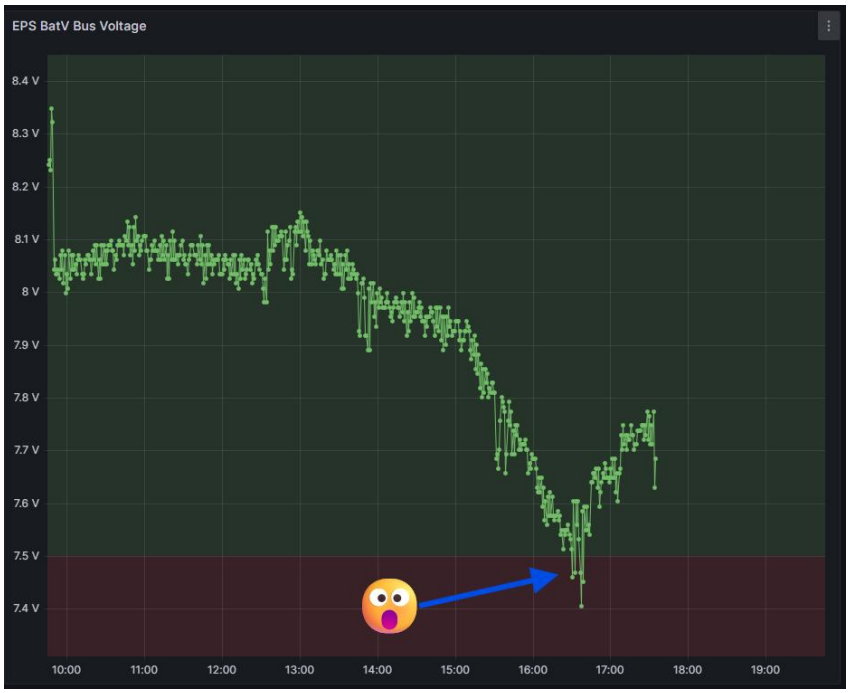


Figure 10. Screenshot from Grafana displaying EPS Battery Voltage dropping to critically power which induced 'safe mode' switch.

PASS 5: 15:30:00 – 15:40:00 UTC

My Role	Spacecraft Operators Manager
Pre-Pass Preparation	Before the pass, the team reviewed the safe mode event from Pass 4 and prepared recovery procedures to return the spacecraft to commissioning mode. Telemetry priorities and payload recovery objectives were also reviewed before AOS.
Specific Objectives of Pass	The main objective of Pass 5 was to recover the spacecraft from safe mode, restore nominal operations and continue downlinking telemetry and EMOD payload data. Finally EMOD was to be deactivated before the end of the pass at LOS.
Pass Length	2026-04-16 16:30:55 - 2026-04-16 16:40:00 Duration: 9 minutes 5 seconds
Procedures Completed During Pass	The communications pass was successfully initialised and spacecraft recovery procedures were carried out during the pass. The spacecraft was successfully returned from safe mode back into to commissioning mode and telemetry monitoring continued until LOS. • **EIR-OPS-002: Pre-Communication Pass Checklist** • **EIR-OPS-003: Start a Communication Pass** • **EIR-OPS-011: Downlink Data From Storage** • **EIR-OPS-022: EMOD Configuration** • **EIR-OPS-025: Safe Mode Entered**
Objectives Achieved	The spacecraft was successfully recovered from safe mode and nominal communications were restored. EVENT, HOUSEKEEPING and EMOD telemetry were successfully downlinked during the pass, confirming healthy payload operation despite the PASS 4 voltage drop anomaly that caused significant problems.
Difficulties Encountered	Recovery operations following the safe mode event required careful coordination between spacecraft monitoring and telemetry downlink activities during the short communications window. As such the time constraint meant the final deactivation of EMOD could not be fully completed.
Data Downlinked	○ EVENT telemetry; `Events_20260416_1730.dat` ○ HOUSEKEEPING telemetry; `HK_20260416_1730.dat` ○ EMOD telemetry; `EMOD_20260416_1730.dat`
Data Uplinked	○ Telecommands associated with spacecraft recovery procedures and restoration of commissioning mode operations.

4/16/2026 16:30	271	INFO	4301	mission.csLEIRSATModeManager.ExitModeSuccess	Record the successful exit of a specified mode.	mission.ModeManager.events	00 00 00 04
4/16/2026 16:30	272	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command from the OBC to the MSP. 0xFF indicates that the MSP has registered a 'failure' to perform some action following a command from the OBC.	platform.ADM.msp	00 03 4b c1
4/16/2026 16:30	273	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command from the OBC to the MSP. 0xFF indicates that the MSP has registered a 'failure' to perform some action following a command from the OBC.	platform.ADM.msp	00 03 4b c1
4/16/2026 16:30	274	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command from the OBC to the MSP. 0xFF indicates that the MSP has registered a 'failure' to perform some action following a command from the OBC.	platform.ADM.msp	00 03 4b c1
4/16/2026 16:30	275	INFO	2802	subsys.ADMRevEIRSAT.SecBurn	Raised when the BurnSecResists action is invoked.	platform.ADM.msp	00 00 00 00
4/16/2026 16:30	276	INFO	4200	mission.csLEIRSATSeparationSequence.StateFunctionComplete	Record that a given state function has completed. State number will go into arguement	mission.SeparationSequence.events	00 00 00 42
4/16/2026 16:30	277	INFO	4302	mission.csLEIRSATModeManager.EnterModeSuccess	Record the successful entry to a specified mode.	mission.ModeManager.events	00 00 00 01
4/16/2026 16:30	278	INFO	319	mission.csLADCSManager.ModeStateSuccess	Raised when the ADCS mode and state are successfully set	mission.ADCSManager.adcs	00 00 00 00
4/16/2026 16:34	279	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 16:35	280	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 16:37	281	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
4/16/2026 16:38	282	WARNING	220	tmtc.TMTCTransfer.GetError	An error occurred whilst processing a get operation	cdh.tmtc.TMTCTransfer.error	00 00 65 96
4/16/2026 16:45	283	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00

Figure 11. Event log extract for Pass 5 taken from full .csv-Excel file of 'EVENTS' data from full Mission Test Day.

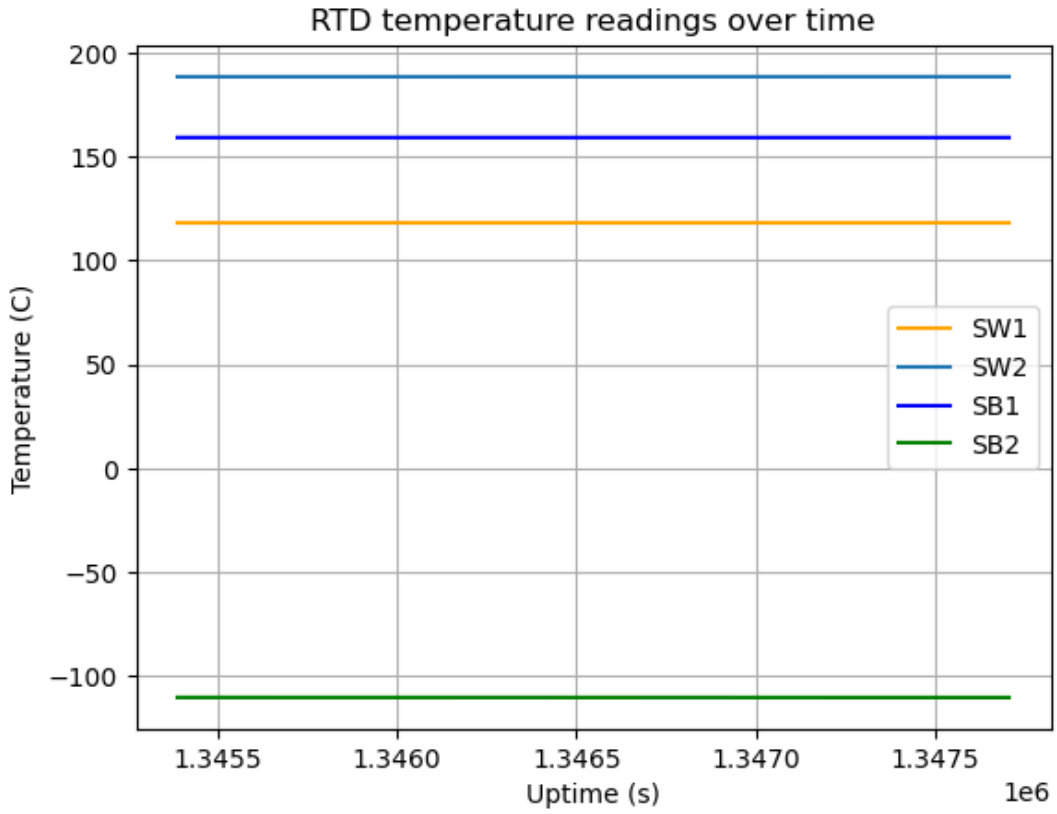


Figure 12. EMOD Telemetry Data configurations for RTD temperature readings over time. Using the .pynb code provided in the Appendix.

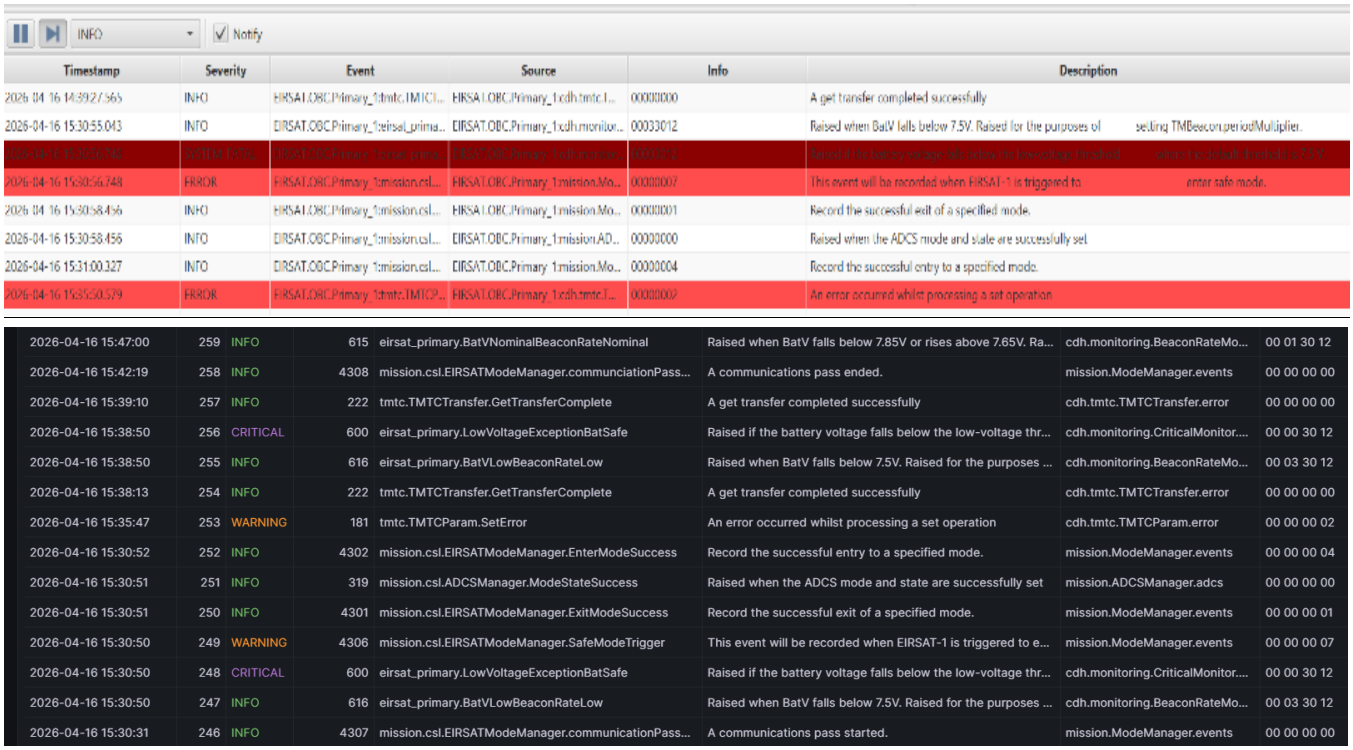
'Never try to hide what you believe may have been a mistake you made…'

One area where I believe I could have performed better was during the final pass while acting as Spacecraft Operators Manager. Although the spacecraft was successfully recovered from safe mode and communications remained stable, I dont think I prepared well enough for the operational procedures required during the pass. As a result, I was slower and not as reliable than I should have been when relaying procedure instructions to the operators during such a limited communications

window. Because of this lack of preparation, I believe I heavily share responsibility for the team not being able to fully complete all planned pass objectives, particularly the deactivation of the EMOD payload before the end of the pass. This pass for me highlighted the importance of being fully familiar with operational procedures before AOS, especially during time-critical recovery operations where efficient communication and coordination are essential.

I also believe I could have responded better during Pass 4 while acting as Daily Pass Planner. During the pass, I noticed the EPS battery voltage warnings (see screenshot below) and gradual voltage drop on the secondary monitoring screen, however I do not think I communicated the issue to the operators quickly enough. In hindsight, earlier escalation of the warning signs may have allowed the SpaCom and spacecraft operators to react sooner and potentially begin recovery procedures earlier before the spacecraft transitioned fully into safe mode. This experience reinforced the importance of continuous situational awareness across all monitoring systems during spacecraft operations, even when not directly acting as an operator. It also highlighted how critical rapid communication and anomaly reporting are during time-sensitive operational situations.

Despite both of these mistakes, I believe the mission test was still highly valuable from an operational perspective. The experience demonstrated the importance of teamwork, preparation, communication and decision-making during live spacecraft operations, particularly when dealing with unexpected problems and limited communications windows.



Timestamp	Severity	Event	Source	Info	Description
2026-04-16 14:59:27.565	INFO	EIRSAT.OBC.Primary_1:tmtc.TMTC1...	EIRSAT.OBC.Primary_1:cdh.tmtc.T...	00000000	A get transfer completed successfully
2026-04-16 15:30:55.043	INFO	EIRSAT.OBC.Primary_1:eirsat_prima...	EIRSAT.OBC.Primary_1:cdh.monitor...	00033012	Raised when BatV falls below 7.5V. Raised for the purposes of setting TMBeaconperiodMultiplier.
2026-04-16 15:30:56.748	ERROR	EIRSAT.OBC.Primary_1:mission.csl...	EIRSAT.OBC.Primary_1:mission.Mo...	00000007	This event will be recorded when EIRSAT-1 is triggered to enter safe mode.
2026-04-16 15:30:58.456	INFO	EIRSAT.OBC.Primary_1:mission.csl...	EIRSAT.OBC.Primary_1:mission.Mo...	00000001	Record the successful exit of a specified mode.
2026-04-16 15:30:58.456	INFO	EIRSAT.OBC.Primary_1:mission.csl...	EIRSAT.OBC.Primary_1:mission.AD...	00000000	Raised when the ADCS mode and state are successfully set
2026-04-16 15:31:00.327	INFO	EIRSAT.OBC.Primary_1:mission.csl...	EIRSAT.OBC.Primary_1:mission.Mo...	00000004	Record the successful entry to a specified mode.
2026-04-16 15:35:50.529	ERROR	EIRSAT.OBC.Primary_1:tmtc.TMTC2...	EIRSAT.OBC.Primary_1:cdh.tmtc.T...	00000002	An error occurred whilst processing a set operation

2026-04-16 15:47:00	259	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:42:19	258	INFO	4308	mission.csl.EIRSATModeManager.communciationPass...	A communications pass ended.	mission.ModeManager.events	00 00 00 00
2026-04-16 15:39:10	257	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
2026-04-16 15:38:50	256	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage thr...	cdh.monitoring.CriticalMonitor....	00 00 30 12
2026-04-16 15:38:50	255	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes ...	cdh.monitoring.BeaconRateMo...	00 03 30 12
2026-04-16 15:38:13	254	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
2026-04-16 15:35:47	253	WARNING	181	tmtc.TMTCParam.SetError	An error occurred whilst processing a set operation	cdh.tmtc.TMTCParam.error	00 00 00 02
2026-04-16 15:30:52	252	INFO	4302	mission.csl.EIRSATModeManager.EnterModeSuccess	Record the successful entry to a specified mode.	mission.ModeManager.events	00 00 00 04
2026-04-16 15:30:51	251	INFO	319	mission.csl.ADCSManager.ModeStateSuccess	Raised when the ADCS mode and state are successfully set	mission.ADCSManager.adcs	00 00 00 00
2026-04-16 15:30:51	250	INFO	4301	mission.csl.EIRSATModeManager.ExitModeSuccess	Record the successful exit of a specified mode.	mission.ModeManager.events	00 00 00 01
2026-04-16 15:30:50	249	WARNING	4306	mission.csl.EIRSATModeManager.SafeModeTrigger	This event will be recorded when EIRSAT-1 is triggered to e...	mission.ModeManager.events	00 00 00 07
2026-04-16 15:30:50	248	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage thr...	cdh.monitoring.CriticalMonitor....	00 00 30 12
2026-04-16 15:30:50	247	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes ...	cdh.monitoring.BeaconRateMo...	00 03 30 12
2026-04-16 15:30:31	246	INFO	4307	mission.csl.EIRSATModeManager.communicationPass...	A communications pass started.	mission.ModeManager.events	00 00 00 00

Figure 13. Both secondary screen Mission Test and Later Grafana screenshots of warning disaplays from EVENT log of voltage drop.

Differences Between the FlatSat and Flight Model Shifts

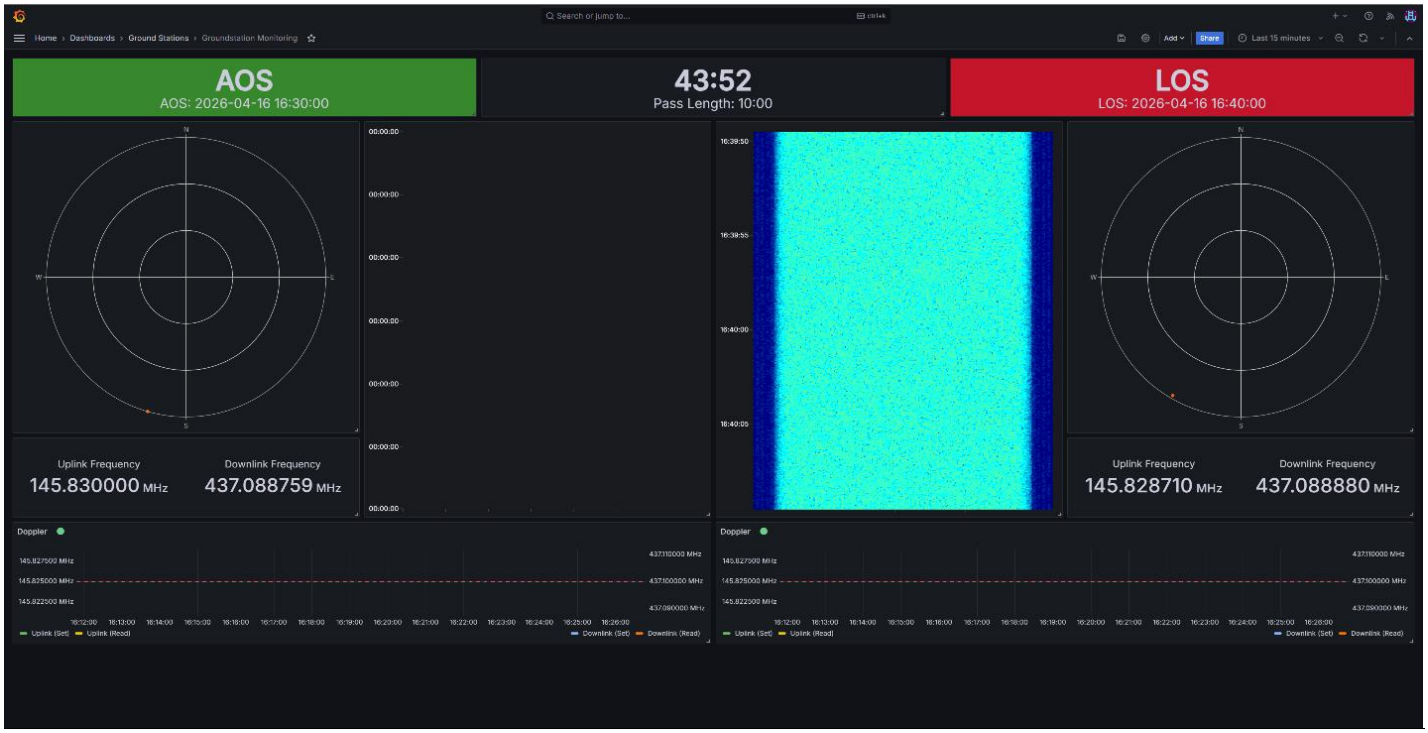


Figure 14. Grafana Ground Station Screenshot including Waterfall Plot, AOS & LOS counter & Pass length duration display.

What were the differences between the FlatSat and Flight Model behaviour?

Even though the mission test used a FlatSat system sitting in a lab instead of an actual spacecraft in orbit, a lot of the operations still felt surprisingly realistic. The combination of RF communications, short communications windows and real operational procedures created genuine pressure (not to mention the grading aspect) during the passes, especially when trying to prioritise telemetry downlinks and complete procedures before LOS. The main difference was obviously the environment itself. Since the FlatSat was operating on a bench in a controlled lab environment, some telemetry such as temperatures remained fairly stable throughout the mission test. In reality, a spacecraft in orbit would experience much larger thermal changes due to sunlight and eclipse periods. However, things like operational mode changes, telemetry downlink and the safe mode event still felt very realistic from an operations perspective.

How effective was the simulation at preparing you for Flight Operations?

Overall, I think the simulation was very effective at preparing us for real flight operations. Before the mission test I underestimated how difficult spacecraft operations could become once limited communications windows, RF constraints and non-nominal scenarios were introduced. Even relatively simple tasks became much harder once there was pressure to complete everything between AOS and LOS. The mission test also gave realistic experience in monitoring telemetry,

following operational procedures and responding to random weird telemetry anomalies in real time. The safe mode event during Pass 4 especially showed how quickly operations can change and how important communication and situational awareness are during spacecraft operations.

Based on the data downlinked, can you identify which aspects of the telemetry give a good impression that the FlatSat might actually be in orbit versus which aspects make it clear that it is really sitting on a bench in a lab?

Some parts of the telemetry genuinely made the mission test feel like we were operating a real spacecraft in orbit. Things like the battery voltage monitoring, operational mode changes, RF communications windows and especially the safe mode transition during Pass 4 made the operations feel very realistic. Having to monitor telemetry in real time through Grafana while working within short communications windows also gave a good impression of what real spacecraft operations are probably like. At the same time, there were some obvious signs that the spacecraft was still sitting on a bench in a lab rather than actually being in orbit. The temperature telemetry stayed fairly stable throughout the mission test and there were no major thermal changes like would normally happen as a spacecraft moves between sunlight and eclipse. The ADCS telemetry also stayed almost completely flat, with gyro rates and angular rate magnitudes showing very little variation during the passes. In reality, a spacecraft in orbit would constantly experience small attitude changes and corrections. The solar cell telemetry was another clear example. In orbit, the solar panels would experience changing sunlight conditions throughout each orbit, so the power generation would vary much more than what we observed during the mission test. Since the FlatSat was operating in a stable lab environment, a lot of this telemetry stayed much more constant than would realistically be expected in space.

How did the introduction of RF communications using ground stations impact the difficulty of operating the spacecraft?

The RF communications aspect made the mission test much more difficult compared to normal bench testing. Having limited communications windows between AOS and LOS added constant time pressure during every pass and meant that telemetry downlink and spacecraft operations had to be prioritised carefully. It also made the mission test feel far more realistic because our operators team couldnt just interact with the spacecraft continuously whenever we needed. Instead, everything had to be prepared in advance and completed during short passes. This became especially difficult during problematic situations like in PASS 4, where operators had to monitor telemetry, communicate with the team and carry out procedures simultaneously while still working within the limited pass duration but with an unexpected curve-ball problematic case thrown in.

3. Part B: Mission Testing and Spacecraft Operations

3. Part B: Mission Testing and Spacecraft Operations

During this mission test I gained a much better understanding of the complexity involved in spacecraft operations and how difficult even the more simple tasks can become once limited communications windows, telemetry constraints and operational procedures are brought into the job. Although the spacecraft used during testing was a FlatSat system rather than a real spacecraft in orbit – thank god!, the use of RF communications, operational procedures and real-time decision making created a very realistic operational environment.

Pass/Contact Planning

Personally, one of the main things I noticed during the mission test was how important pass planning is to successful spacecraft operations. Due to the short communications windows between AOS and LOS, it was not possible to complete every desired operation during a single pass. As a result, things like telemetry downlinking, payload operations and spacecraft recovery procedures had to be prioritised carefully before each pass. During my role as Daily Pass Planner, I also realised how important situational awareness is when planning operations, particularly when spacecraft power levels and operational modes can change unexpectedly during a pass.

Orbits

Although the FlatSat was located within a laboratory environment, the mission test still simulated several realistic orbital operational constraints. The limited duration of communications passes created a realistic representation of spacecraft visibility windows from the horizon during each orbit. The scheduled passes and RF acquisition times created a lot of added time pressure throughout operations and demonstrated how orbital access really limits when operators can actually communicate with and control the spacecraft.

Spacecraft Communication

The introduction of RF communications significantly increased the difficulty of operating the spacecraft compared to direct bench testing. Operators were required to wait for AOS before communication could begin and all operations had to be completed before LOS occurred. Communication delays, telemetry prioritisation and limited pass durations made even routine tasks more challenging. The use of operational procedures such as EIR-OPS-002, EIR-OPS-003 and EIR-OPS-006 also reinforced the importance of following strict operational workflows during spacecraft operations.

Spacecraft Control and Telemetry

The mission test provided realistic experience in monitoring spacecraft telemetry and responding to spacecraft state changes in real time. Throughout the mission test, Grafana telemetry dashboards and housekeeping telemetry were used to assess spacecraft health, including monitoring battery voltage, temperatures, operational modes and event activity. One of the most important operational events occurred during Pass 4 when the spacecraft battery voltage dropped and the spacecraft entered safe mode. This demonstrated how important telemetry monitoring and rapid operator communication are during problem based situations. It also showed how spacecraft control

decisions are heavily dependent on accurate telemetry interpretation during short communications windows.

Data Handling

Data handling was another huge operational challenge during the mission test. Because the spacecraft had stored one week of telemetry onboard, operators had to prioritise what telemetry should be downlinked during each communications pass. EVENT, HOUSEKEEPING, EMOD and TED telemetry all competed for limited downlink time, meaning careful planning was required throughout operations. The use of Large Data Transfers (LDTs), telemetry row selection and storage downlink procedures demonstrated how spacecraft data management becomes increasingly important when communications bandwidth and contact duration are limited.

Experience as a Test Support Operator

During the mission test I also acted as a Test Support Operator. This role was important because the Test Support Operators monitored the progress of spacecraft operations, coordinated support procedures and helped maintain the as-run Mission Test documentation throughout operations. The role also involved monitoring the state of the ground support equipment and supporting the spacecraft operators when required.

From my experience, the Test Support Operator role played a major part in ensuring the mission test ran efficiently. The support operators helped reduce operator workload during time-critical communications windows and assisted with maintaining operational awareness during spacecraft activities. During scary situations, particularly the safe mode event during Pass 4, support operator coordination became even more important as multiple activities and telemetry sources needed to be monitored simultaneously.

4. Summary / Conclusions

Overall, I believe the mission test was successful despite not all mission objectives being fully completed. The exercise provided a very realistic experience of spacecraft operations and demonstrated how challenging even routine activities can become once RF communications constraints, limited pass durations and real-time operational decision making are introduced. Throughout the mission test, the team successfully carried out multiple communications passes involving pre-pass preparation, real-time spacecraft operations, telemetry monitoring and post-pass analysis.

A large quantity of EVENT, HOUSEKEEPING, EMOD and TED telemetry was successfully downlinked and used to assess spacecraft and payload health throughout the mission test. The team also successfully carried out EMOD payload operations, telemetry recovery and spacecraft recovery procedures following the safe mode event during Pass 4. In addition, Large Data Transfer (LDT) files containing EMOD telemetry were successfully recovered and decoded, allowing EMOD temperature data to be analysed against mission elapsed time.

One of the most valuable parts of the mission test in my opinion was actually experiencing how operational pressure affects spacecraft decision making during short communications windows. The introduction of RF communications and ground station limits made spacecraft operations far more difficult compared to normal bench testing and really showed the importance of real pre-mission preparation, team communication and situational awareness during live operations. The safe mode scare during Shift 2 also really showed us how important telemetry monitoring and rapid operator coordination resposes are during contingency situations.

Although some planned objectives were not fully completed, particularly during the final recovery pass where the EMOD payload was not successfully deactivated before LOS, I still believe the mission test achieved its overall purpose very effectively. The exercise provided realistic operational experience, improved our familiarity with mission control systems and operational procedures, and highlighted both the technical and human challenges associated with spacecraft operations. Most importantly, the mission test showed how valuable mistakes and unexpected problematic sabotages can be from a learning perspective, particularly in improving operational awareness, teamwork and preparation for future spacecraft operations.

5. Appendix

BETA Briefing Notes

- **Our Main Mission Objective:** EMOD Degradation - Downlink & review logged data
- **Notes on launch and review.**
  - Successful launch
  - Commissioning work needs to be completed
  - Micro-controller double duty as ADM, ADM is off, programme EMOD to download.
  - Its been running 1 week in orbit, there will be lots of data, we want most recent data

- **PROCEDURES;**

1. OPS.SIM@ _ , - Has to be accessed via hotspot as Eduroam and UCD wifi wont work. Needed; 'PRE-COMMS', 'START A PASS',
2. 'EIROPS 002', 'EIROPS 003' for every pass, downlink data from storage.
3. 1ST Procedure; EIROPS 006.7 – EMOD Reprogramming
4. Order of OPS; 006.7, 006.8 – Health Check, 022 – Data Run.
5. SECTION A; To check basic settings.
6. SECTION B; To ensure mode 0; Step B.4 says to jump to A.9, This is a mistake, don't do this.
7. Set all RTDs enabled using 'Section C' (RTD Sampling settings)
8. Use all of them, don't 'ignore' any of the RTDs – ENABLE ALL
9. SECTION D; Sampling rate @ 5s.
10. Jump back to section B, mode 1.
11. Once in mode 1, SECTION E; Set OBC to 'get' data from EMOD & log, AUTOPAGES, AUTOPOLL.
12. At this point its up and running, We will use all. Run it for one full orbit.
13. After one full orbit, use EIROPS SECTION 2F, to turn off EMOD.
14. Throughout, should be downlinking data. With main priority given to Channel 8; Events. Thereafter, priority for health checks; Channel 15: Housekeeping. Final priority given to Channel 16: TED & 19: Power systems. Then EMOD, on Channel 77.

- **NEEDED;**

1. De-aggregator; write a decoder for EMOD data. If it works correctly it should look like the curve PDF on Brightspace.
2. TS & TP or TSTP. (Separate is more traditional)
3. FOR TS;
   a. Hit the bullet points; Requirements – Pick appropriate for verification, Doc 'Test Requirements' can be used as inspiration. We have flat-sat not assembled. We are in operations room. We can plagiarize but it must actually apply to us. Test Organization; PASS/FAIL Criteria general overall, Test Operators & Roles should be abstract as no one role each.
   b. Passes; 5x 10mins approx. over 2 shifts; S1has 2 passes, S2 has 3 passes. All will be UTC; 1ST @ 11:45-11:55, 2ND @ 12:45-12:55 , 3RD 14:30-14:40, 4TH @ 15:30-15:40, 5TH 16:30-16:40
   c. Scenarios will be tested '*Scenario 10: Payload Reprogramming*' & '*Scenario 7: Data Downlink*'. Description of each scenario; Objectives, Procedures are given.
4. FOR TP; Safe to copy directly for Brightspace. But write-up has to be for our scenarios; Objectives and P/F criteria. Must reference OPS manual.

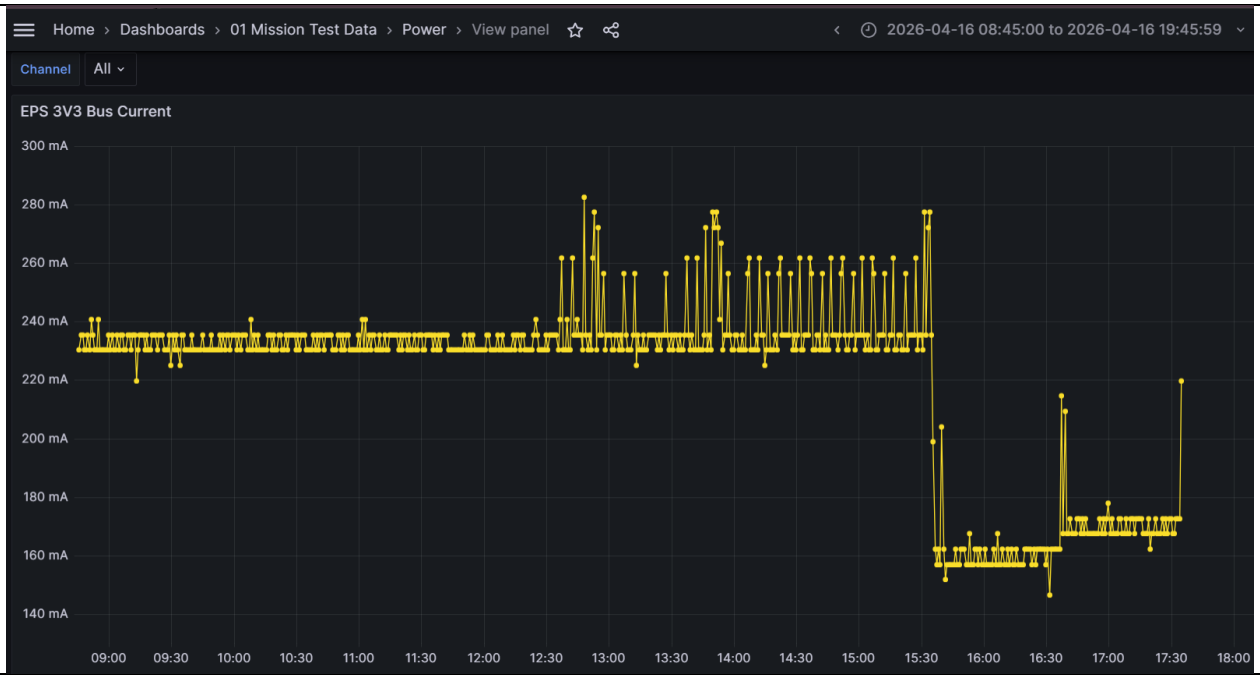
Full Day (07:45:00 – 18:45:00 UTC+2) Grafana Screenshots




Home › Dashboards › 01 Mission Test Data › Housekeeping
2026-04-16 07:45:00 to 2026-04-16 18:45:59
EPS Bus Currents
BatV 3.3V 5V 12V
Battery
Battery BatV Bus Voltage
Current from Battery
Battery Cell Temperatures
Battery Cell Heater Status
Cell Pair 1
Cell Pair 2
Cell Pair 3
Off
Home › Dashboards › 01 Mission Test Data › Housekeeping › View panel
Channel
All
Events
EPS BatV Bus Voltage
2026-04-16 16:37:09
BatV 7.41 V

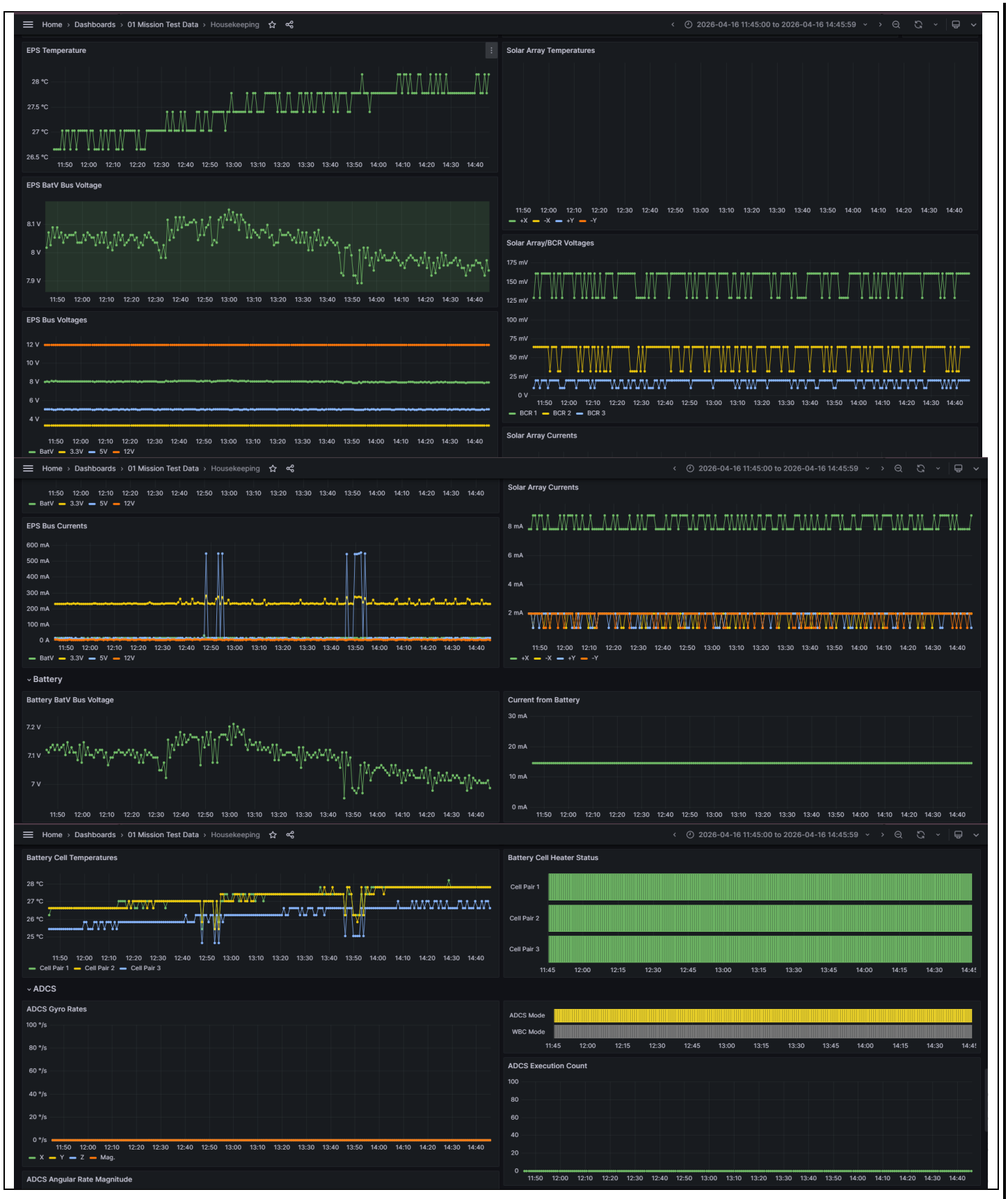

Home › Dashboards › 01 Mission Test Data › Power › View panel
2026-04-16 09:45:00 to 2026-04-16 20:45:59
Channel
All
EPS Bus Currents
600 mA
550 mA
500 mA
450 mA
400 mA
350 mA
300 mA
250 mA
200 mA
150 mA
100 mA
50 mA
0 A
10:00 10:30 11:00 11:30 12:00 12:30 13:00 13:30 14:00 14:30 15:00 15:30 16:00 16:30 17:00 17:30 18:00 18:30 19:00
BatV 3V3 5V

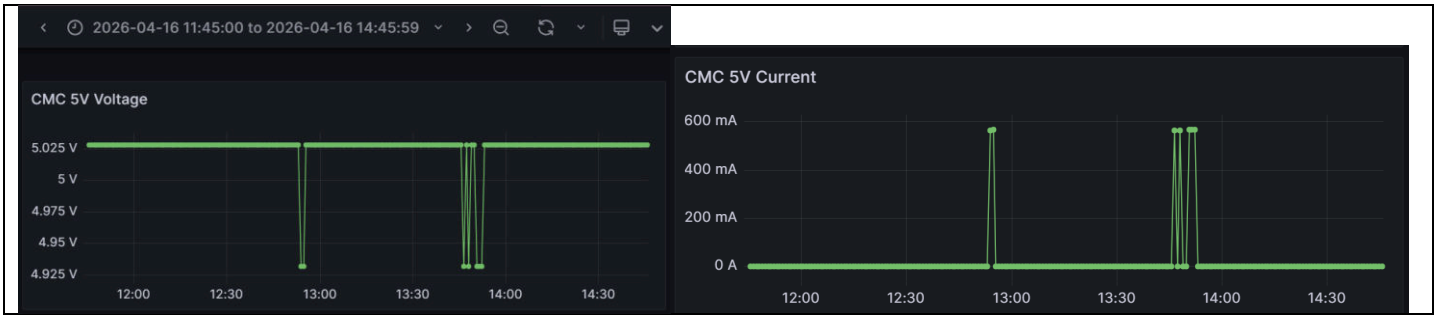

Home › Dashboards › 01 Mission Test Data › Power › View panel
2026-04-16 12:00:00 to 2026-04-16 19:45:59
Channel
All
CMC 5V Voltage
5.03 V
5.02 V
5.01 V
5 V
4.99 V
4.98 V
4.97 V
4.96 V
4.95 V
4.94 V
4.93 V
12:00 12:30 13:00 13:30 14:00 14:30 15:00 15:30 16:00 16:30 17:00 17:30 18:00 18:30 19:00 19:30

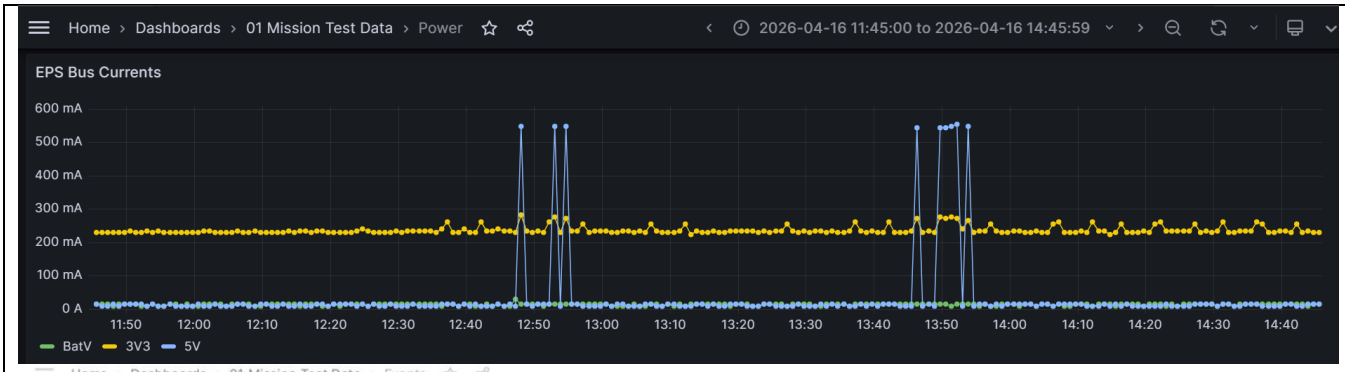

Home › Dashboards › 01 Mission Test Data › Power › View panel
2026-04-16 08:45:00 to 2026-04-16 19:45:59
Channel
All
EPS 3V3 Bus Current
300 mA
280 mA
260 mA
240 mA
220 mA
200 mA
180 mA
160 mA
140 mA
09:00
09:30
10:00
10:30
11:00
11:30
12:00
12:30
13:00
13:30
14:00
14:30
15:00
15:30
16:00
16:30
17:00
17:30
18:00

SHIFT 1 (11:45:00 – 14:45:00 UTC+2) Grafana Screenshots




Home › Dashboards › 01 Mission Test Data › Housekeeping
2026-04-16 11:45:00 to 2026-04-16 14:45:59
EPS Temperature
Solar Array Temperatures
EPS BatV Bus Voltage
Solar Array/BCR Voltages
EPS Bus Voltages
Solar Array Currents
EPS Bus Currents
Battery
Battery BatV Bus Voltage
Current from Battery
Battery Cell Temperatures
Battery Cell Heater Status
ADCS
ADCS Gyro Rates
ADCS Mode
WBC Mode
ADCS Execution Count
ADCS Angular Rate Magnitude
BatV 3.3V 5V 12V
+X -X +Y -Y
BCR 1 BCR 2 BCR 3
Cell Pair 1 Cell Pair 2 Cell Pair 3
X Y Z Mag.


2026-04-16 11:45:00 to 2026-04-16 14:45:59
CMC 5V Voltage
5.025 V
5 V
4.975 V
4.95 V
4.925 V
12:00
12:30
13:00
13:30
14:00
14:30
CMC 5V Current
600 mA
400 mA
200 mA
0 A
12:00
12:30
13:00
13:30
14:00
14:30

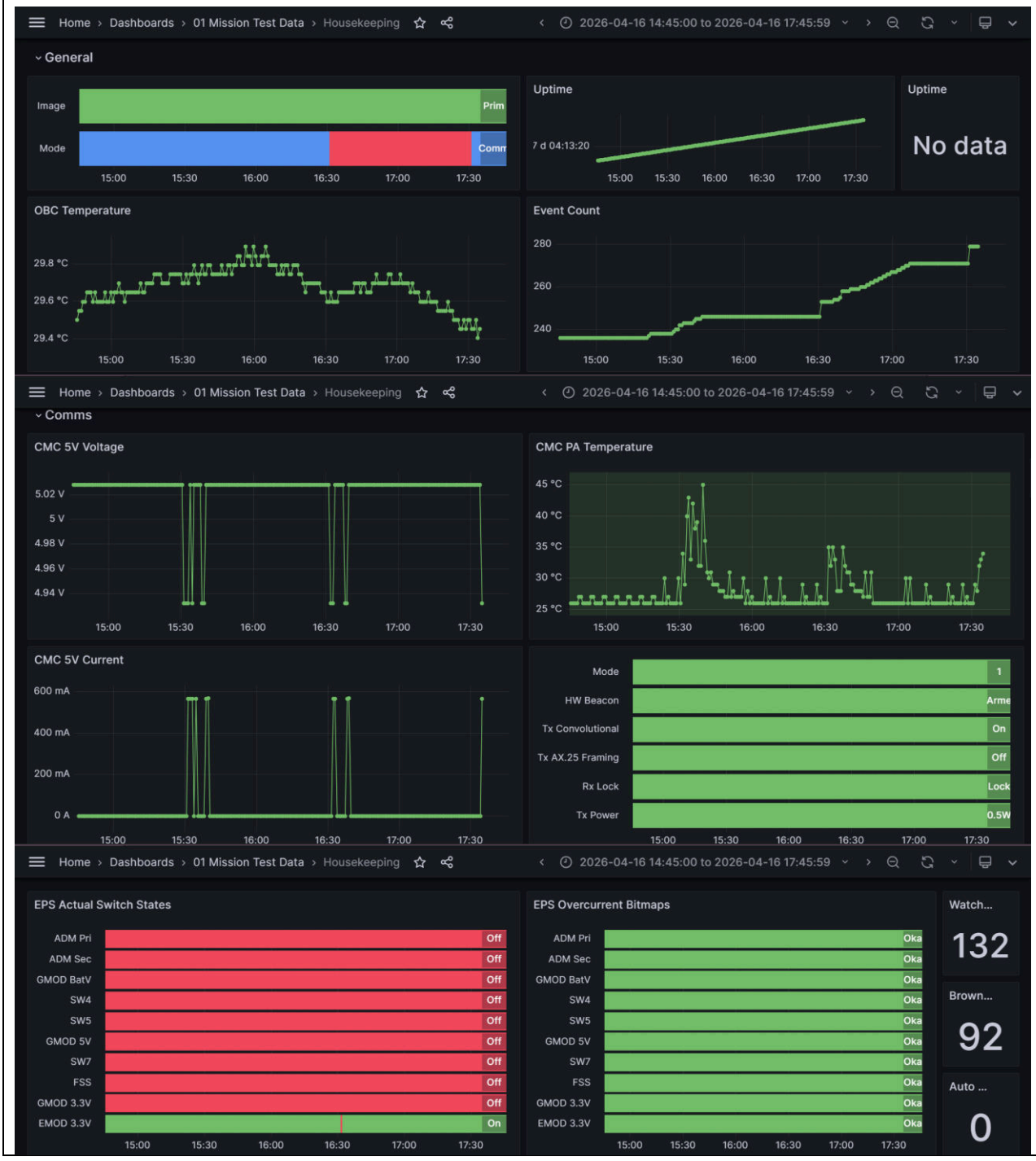


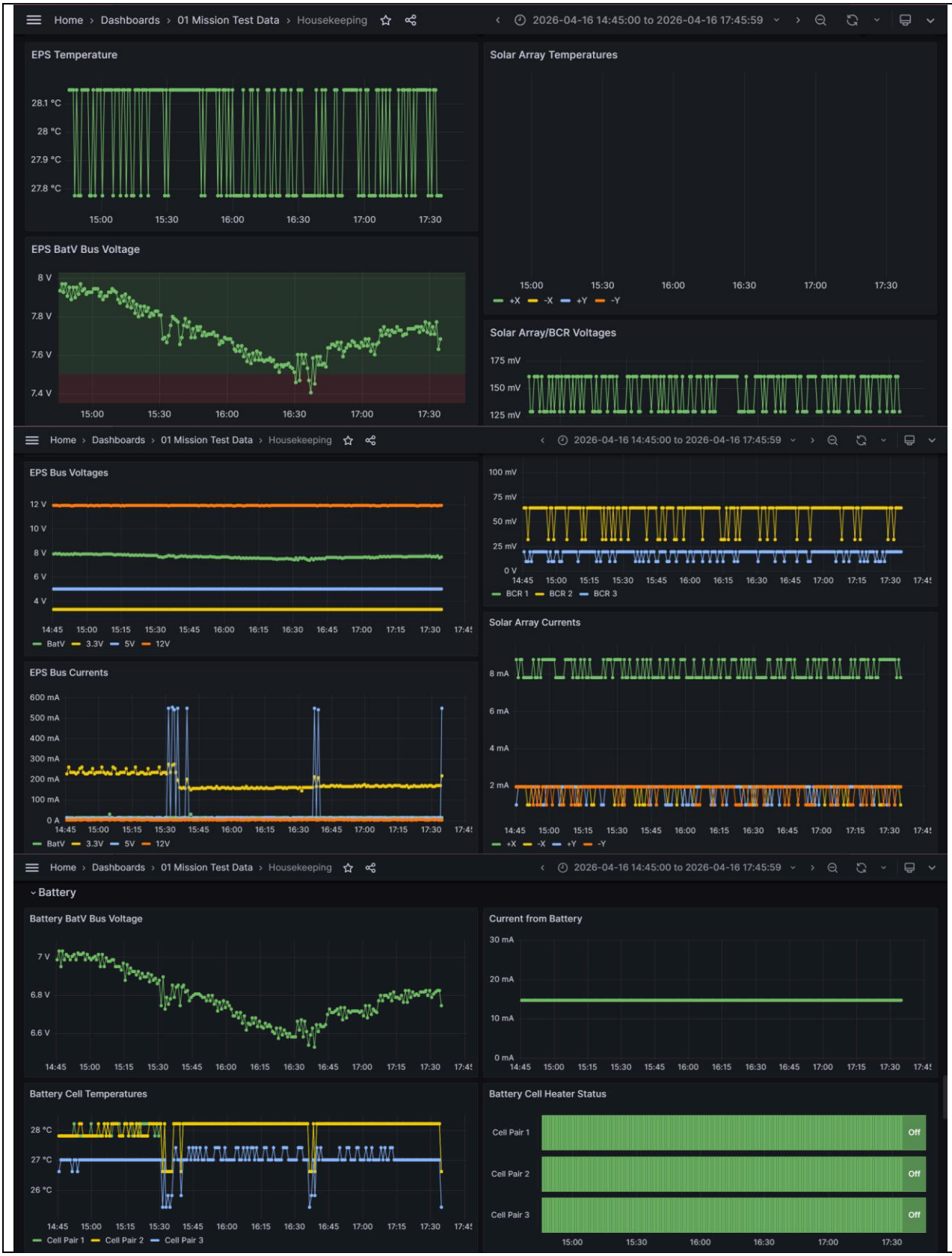
Home › Dashboards › 01 Mission Test Data › Events

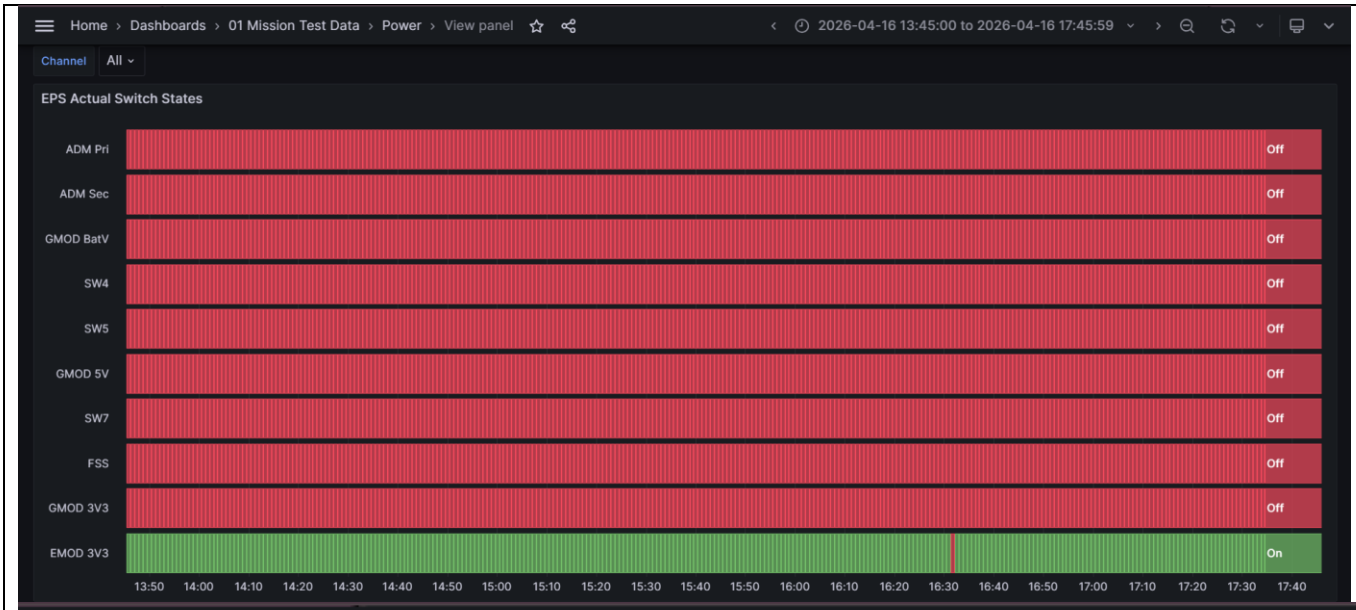
Events

Time	ARL	Level	Event ID	Event Name	Description
2026-04-16 14:42:29	245	INFO	4308	mission.csl.EIRSATModeManager.communciationPass...	A communications pass ended.
2026-04-16 14:40:01	244	WARNING	223	tmtc.TMTCTransfer.GetTransferError	A get transfer reported an error, the transfer did not comple
2026-04-16 14:39:21	243	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 14:35:01	242	INFO	2900	subsys.EMODRevB.ReprogrammeComplete	Raised when the EMOD MSP has been reprogrammed.
2026-04-16 14:33:32	241	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 14:33:18	240	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 14:32:09	239	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 14:30:47	238	INFO	4307	mission.csl.EIRSATModeManager.communicationPass...	A communications pass started.
2026-04-16 14:22:00	237	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra
2026-04-16 14:20:40	236	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra
2026-04-16 13:10:31	235	WARNING	121	logging.DataLogger.LogError	An error occurred whilst attempting to log new data to the l
2026-04-16 13:05:20	234	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 13:03:30	233	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 13:01:40	232	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 13:00:00	231	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 12:58:20	230	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 12:57:19	229	INFO	4308	mission.csl.EIRSATModeManager.communciationPass...	A communications pass ended.
2026-04-16 12:56:10	228	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 12:55:10	227	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose.
2026-04-16 12:54:13	226	WARNING	190	tmtc.TMTCAction.InvokeError	An error occurred whilst processing an invoke operation
2026-04-16 12:54:03	225	WARNING	190	tmtc.TMTCAction.InvokeError	An error occurred whilst processing an invoke operation
2026-04-16 12:53:48	224	WARNING	180	tmtc.TMTCParam.GetError	An error occurred whilst processing a get operation
2026-04-16 12:52:36	223	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 12:52:08	222	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 12:51:30	221	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 12:50:52	220	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 12:49:43	219	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully
2026-04-16 12:49:00	218	INFO	614	eirsat_primary.BatVHighBeaconRateHigh	Raised when BatV rises above 8.0V. Raised for the purpose...
2026-04-16 12:47:49	217	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully

SHIFT 2 (14:45:00 – 17:45:00 UTC+2) Grafana Screenshots




Home › Dashboards › 01 Mission Test Data › Housekeeping
2026-04-16 14:45:00 to 2026-04-16 17:45:59
EPS Temperature
28.1 °C
28 °C
27.9 °C
27.8 °C
15:00 15:30 16:00 16:30 17:00 17:30
Solar Array Temperatures
15:00 15:30 16:00 16:30 17:00 17:30
+X -X +Y -Y
EPS BatV Bus Voltage
8 V
7.8 V
7.6 V
7.4 V
15:00 15:30 16:00 16:30 17:00 17:30
Solar Array/BCR Voltages
175 mV
150 mV
125 mV
Home › Dashboards › 01 Mission Test Data › Housekeeping
2026-04-16 14:45:00 to 2026-04-16 17:45:59
EPS Bus Voltages
12 V
10 V
8 V
6 V
4 V
BatV 3.3V 5V 12V
100 mV
75 mV
50 mV
25 mV
0 V
BCR 1 BCR 2 BCR 3
EPS Bus Currents
600 mA
500 mA
400 mA
300 mA
200 mA
100 mA
0 A
BatV 3.3V 5V 12V
Solar Array Currents
8 mA
6 mA
4 mA
2 mA
+X -X +Y -Y
Home › Dashboards › 01 Mission Test Data › Housekeeping
2026-04-16 14:45:00 to 2026-04-16 17:45:59
Battery
Battery BatV Bus Voltage
7 V
6.8 V
6.6 V
Current from Battery
30 mA
20 mA
10 mA
0 mA
Battery Cell Temperatures
28 °C
27 °C
26 °C
Cell Pair 1 Cell Pair 2 Cell Pair 3
Battery Cell Heater Status
Cell Pair 1 Off
Cell Pair 2 Off
Cell Pair 3 Off
15:00 15:30 16:00 16:30 17:00 17:30

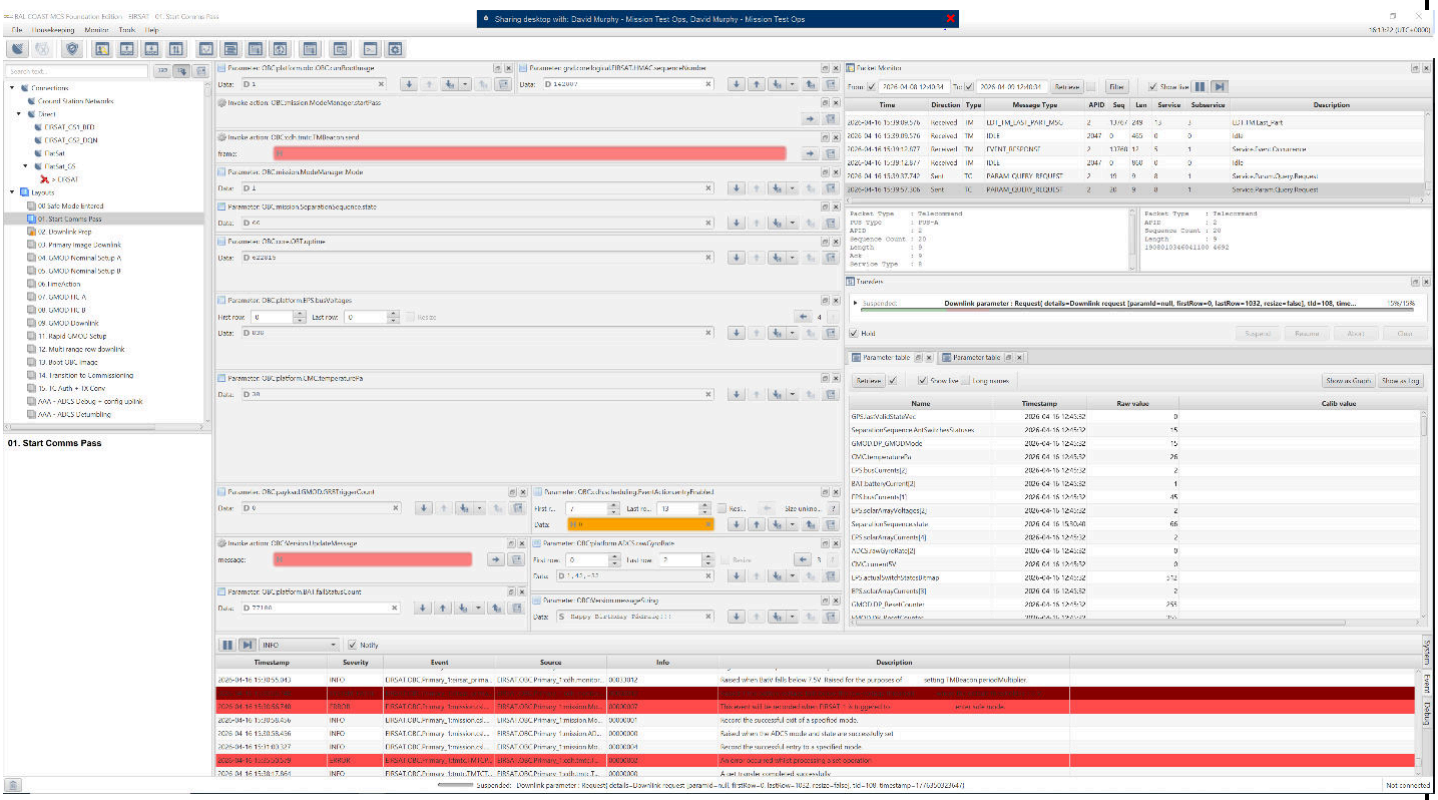


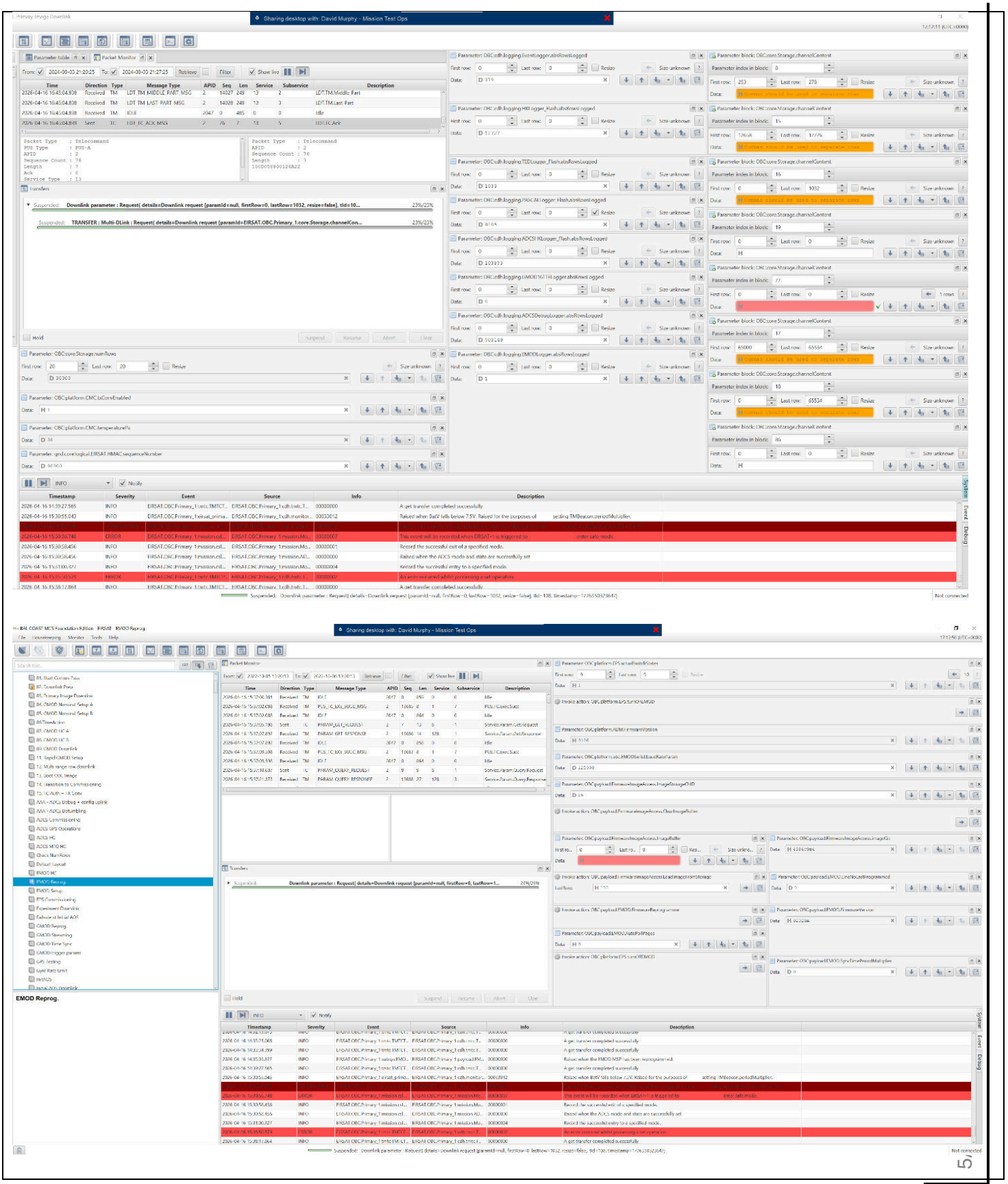
Home › Dashboards › 01 Mission Test Data › Events

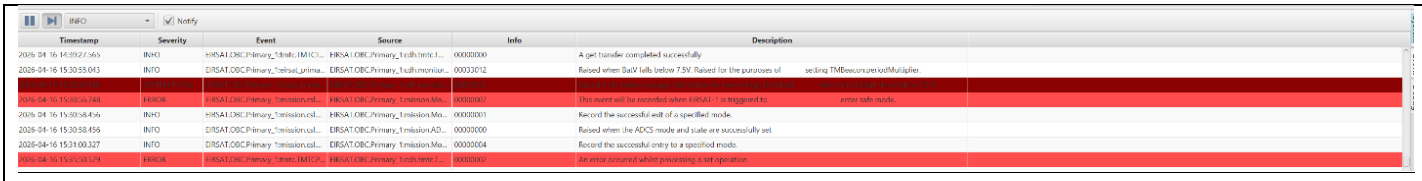
Events

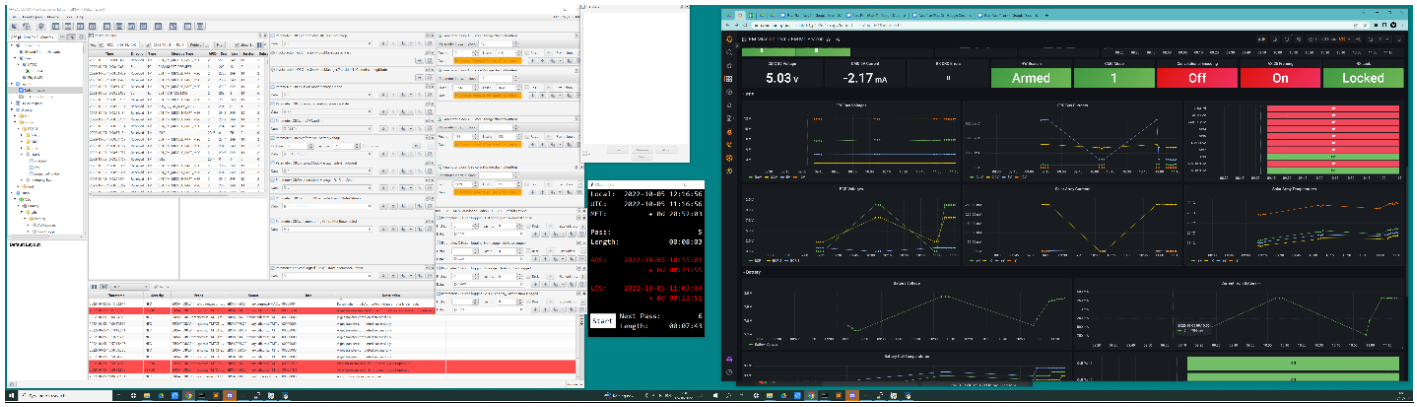
Time	ARL	Level	Event ID	Event Name	Description	Source	Info
2026-04-16 16:30:55	276	INFO	4200	mission.cslEIRSATSeparationSequence.StateFunction...	Record that a given state function has completed. State nu...	mission.SeparationSequence.e...	00 00 00 42
2026-04-16 16:30:55	275	INFO	2802	subsys.ADMRevEIRSAT.SecBurn	Raised when the BurnSecResists action is invoked.	platform.ADM.msp	00 00 00 00
2026-04-16 16:30:55	274	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command fr...	platform.ADM.msp	00 03 4b c1
2026-04-16 16:30:55	273	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command fr...	platform.ADM.msp	00 03 4b c1
2026-04-16 16:30:55	272	INFO	2800	subsys.ADMRevEIRSAT.CommandFailed	Raised when 0xFF is returned in response to a command fr...	platform.ADM.msp	00 03 4b c1
2026-04-16 16:30:55	271	INFO	4301	mission.cslEIRSATModeManager.ExitModeSuccess	Record the successful exit of a specified mode.	mission.ModeManager.events	00 00 00 04
2026-04-16 16:06:40	270	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 16:04:50	269	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 16:03:00	268	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 16:01:20	267	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:59:19	266	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:57:39	265	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:55:59	264	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:54:09	263	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:52:19	262	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:50:39	261	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:49:00	260	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:47:00	259	INFO	615	eirsat_primary.BatVNominalBeaconRateNominal	Raised when BatV falls below 7.85V or rises above 7.65V. Ra...	cdh.monitoring.BeaconRateMo...	00 01 30 12
2026-04-16 15:42:19	258	INFO	4308	mission.cslEIRSATModeManager.communciationPass...	A communications pass ended.	mission.ModeManager.events	00 00 00 00
2026-04-16 15:39:10	257	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
2026-04-16 15:38:50	256	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage thr...	cdh.monitoring.CriticalMonitor...	00 00 30 12
2026-04-16 15:38:50	255	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes ...	cdh.monitoring.BeaconRateMo...	00 03 30 12
2026-04-16 15:38:13	254	INFO	222	tmtc.TMTCTransfer.GetTransferComplete	A get transfer completed successfully	cdh.tmtc.TMTCTransfer.error	00 00 00 00
2026-04-16 15:35:47	253	WARNI...	181	tmtc.TMTCParam.SetError	An error occurred whilst processing a set operation	cdh.tmtc.TMTCParam.error	00 00 00 02
2026-04-16 15:30:52	252	INFO	4302	mission.cslEIRSATModeManager.EnterModeSuccess	Record the successful entry to a specified mode.	mission.ModeManager.events	00 00 00 04
2026-04-16 15:30:51	251	INFO	319	mission.cslADCSManager.ModeStateSuccess	Raised when the ADCS mode and state are successfully set	mission.ADCSManager.adcs	00 00 00 00
2026-04-16 15:30:51	250	INFO	4301	mission.cslEIRSATModeManager.ExitModeSuccess	Record the successful exit of a specified mode.	mission.ModeManager.events	00 00 00 01
2026-04-16 15:30:50	249	WARNI...	4306	mission.cslEIRSATModeManager.SafeModeTrigger	This event will be recorded when EIRSAT-1 is triggered to e...	mission.ModeManager.events	00 00 00 07
2026-04-16 15:30:50	248	CRITICAL	600	eirsat_primary.LowVoltageExceptionBatSafe	Raised if the battery voltage falls below the low-voltage thr...	cdh.monitoring.CriticalMonitor...	00 00 30 12
2026-04-16 15:30:50	247	INFO	616	eirsat_primary.BatVLowBeaconRateLow	Raised when BatV falls below 7.5V. Raised for the purposes ...	cdh.monitoring.BeaconRateMo...	00 03 30 12
2026-04-16 15:30:31	246	INFO	4307	mission.cslEIRSATModeManager.communicationPass...	A communications pass started.	mission.ModeManager.events	00 00 00 00

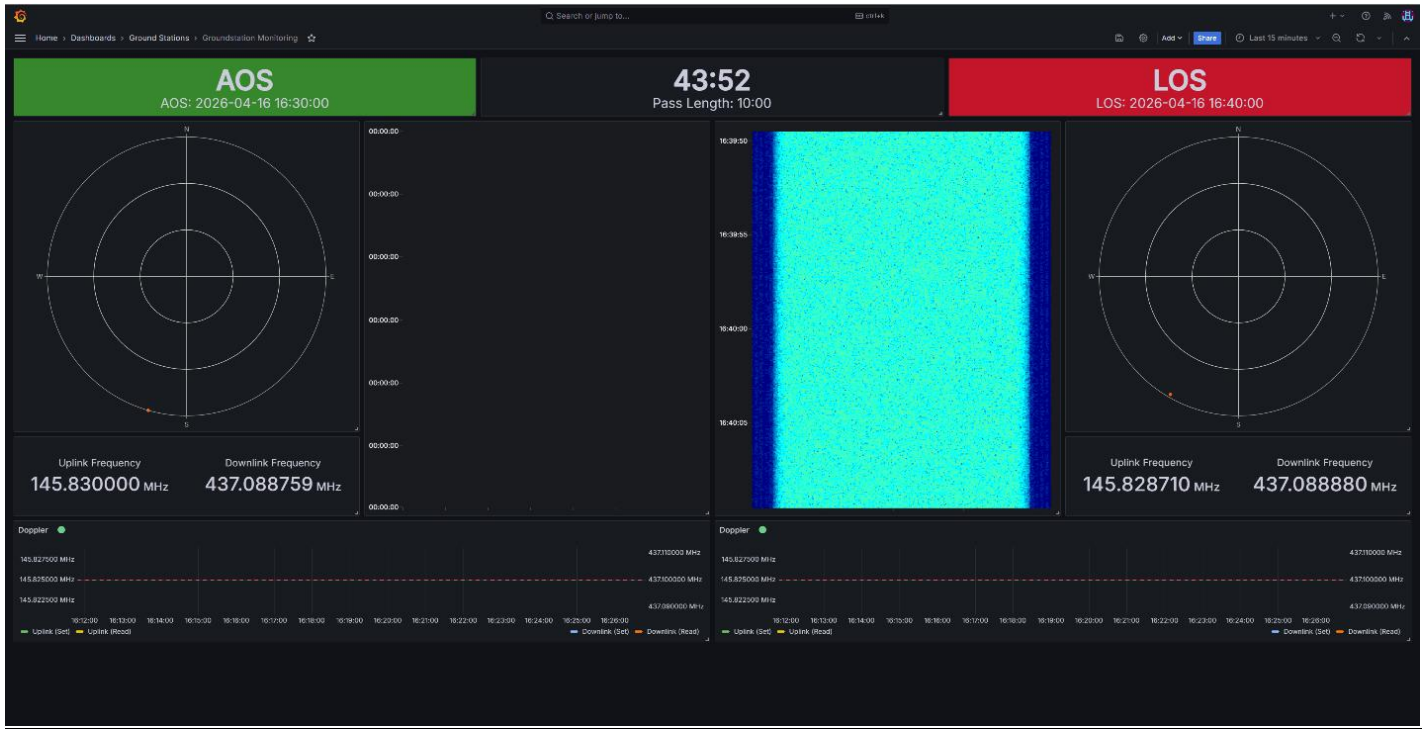
MCS Software Screenshots from Mission Test




Sharing desktop with: David Murphy - Mission Test Ops
EMOD Reprog.


INFO
Notify
Timestamp
Severity
Event
Source
Info
Description
2026-04-16 14:39:27.565
INFO
EIRSAT.OBC.Primary_1.tmtc.TMTC1...
EIRSAT.OBC.Primary_1.cdh.tmtc.T...
00000000
A get transfer completed successfully
2026-04-16 15:30:55.043
INFO
EIRSAT.OBC.Primary_1.eirsat_prima...
EIRSAT.OBC.Primary_1.cdh.monitor...
00033012
Raised when BatV falls below 7.5V. Raised for the purposes of
setting TMBeacon.periodMultiplier.
2026-04-16 15:30:56.748
ERROR
EIRSAT.OBC.Primary_1.mission.csl...
EIRSAT.OBC.Primary_1.mission.Mo...
00000007
This event will be recorded when EIRSAT-1 is triggered to
enter safe mode.
2026-04-16 15:30:58.456
INFO
EIRSAT.OBC.Primary_1.mission.csl...
EIRSAT.OBC.Primary_1.mission.Mo...
00000001
Record the successful exit of a specified mode.
2026-04-16 15:30:58.456
INFO
EIRSAT.OBC.Primary_1.mission.csl...
EIRSAT.OBC.Primary_1.mission.AD...
00000000
Raised when the ADCS mode and state are successfully set
2026-04-16 15:31:00.327
INFO
EIRSAT.OBC.Primary_1.mission.csl...
EIRSAT.OBC.Primary_1.mission.Mo...
00000004
Record the successful entry to a specified mode.
2026-04-16 15:55:50.579
ERROR
EIRSAT.OBC.Primary_1.tmtc.TMTC1...
EIRSAT.OBC.Primary_1.cdh.tmtc.T...
00000007
An error occurred whilst processing a set operation


5.03 V
-2.17 mA
Armed
1
Off
On
Locked


Home › Dashboards › Ground Stations › Groundstation Monitoring
AOS
AOS: 2026-04-16 16:30:00
43:52
Pass Length: 10:00
LOS
LOS: 2026-04-16 16:40:00
Uplink Frequency
145.830000 MHz
Downlink Frequency
437.088759 MHz
Uplink Frequency
145.828710 MHz
Downlink Frequency
437.088880 MHz
Doppler
Doppler